SMART ATTEND: RFID ENABLED ATTENDANCE SYSTEM FOR AUTOMATED EMPLOYEE OR STUDENT TRACKING

A project report is submitted in partial fulfilment of the requirements for the award and degree of

BACHELOR OF TECHNOLOGY

With specialization in

ELECTRONICS AND COMMUNICATION ENGINEERING

Submitted by

K. Srinadh	-	21F91A04B0
S.Ganesh	-	21F91A04C7
G.Parthasaradahi reddy	-	21F91A0478
R.Tharun	-	21F91A04B8
Ch.Teja	-	21F91A04B6

Under the esteemed guidance of

Mr. I. Rama Koteswara Rao, M.Tech

Associate Professor

DEPARTMENT OF

ELECTRONICS AND COMMUNICATON ENGINEERING

PRAKASAM ENGINEERING COLLEGE

(App by AICTE, New Delhi & Affiliated to JNTUK, KAKINADA)

Kandukur-OV Road ,523105, Prakasam A.P.

2021-2025

PRAKASAM ENGINEERING COLLEGE

(App by AICTE, New Delhi & Affiliated to JNTUK, KAKINADA)

Kandukur-OV Road ,523105, Prakasam A.P.

2021-2025

DEPARTMENT OF

ELECTRONICS AND COMMUNICATON ENGINEERING

CERTIFICATE

This is to certity that the project entitled **"Smart Attend: RFID-Enabled Attendance System For Automated Employee or Student Tracking"** is a Bonafide work of K.Srinadh – 21F91A04B0, S.Ganesh - 21F91A04C7, G.Parthasaradhi Reddy – 21F91A0478, R.Tharun – 21F91A04B8, Ch.Teja – 21F91A04B6. In the partial fulfilment Of the requirements for the award and degree of **BACHELOR OF TECHNOLOGY** With specialization in ELECTRONICS AND COMMUNICATION ENGINEERING in the Academic year 2024-2025. This is done under the supervision and guidance.

Project Guide

Mr. I. Rama Koteswara Rao, M.Tech
Associate Professor

Head Of The Department

Dr. G.Suresh, HOD
Professor

SIGNATURE OF INTERNAL EXAMINER

SIGNATURE OF EXTERNAL EXAMINER

ACKNOWLEDGEMENT

A successful project is a fruitful culmination of efforts by many people, some members were Directly involved and some other members were quickly encouraged and supported from the background. We owe a great thanks to the great people who helped and supported us during the accomplishment of the project. Our endeavour stands unaccomplished without gratitude to persons allot towards the success of our project work. We are grateful thanks to our guide **I. Rama Koteswara rao**, **Assoc. professor**, under whose guidance the project work is carried out. We are thankful **Dr. G. Suresh, Assoc. professor and HOD, Dept of ECE** for being a full source of inspiration throughout our study in this college. We are thankful to all teaching and non-teaching staff and our parents for their encouragement and support at every stage of this work. We would like to thank all those who have directly or indirectly helped in the accomplishment of this project. Finally, we acknowledge sincerely the effective services rendered by one and all involved directly and indirectly in the entire project.

Submitted by:

K.Srinadh

S. Ganesh

G. Parthasaradhi

R. Tharun

Ch. teja

ABSTRACT

RFID (Radio Frequency Identification) technology is an innovative and automated wireless identification system that operates using a combination of RFID tags (cards) and a reader. The tags, which come in two types—active and passive—transmit data to the reader, which then processes the information. RFID technology has gained significant attention in various fields due to its ability to streamline processes by automating identification and tracking, thus eliminating the need for manual intervention.

One of the primary applications of RFID technology is in the area of attendance management. Traditional methods of recording attendance, such as roll calls or sign-in sheets, are time-consuming, prone to errors, and can be inefficient. This study aims to address these challenges by utilizing RFID technology to automate the attendance process, providing a more organized, efficient, and time-saving solution.

In this system, each individual is assigned an RFID card (tag) that contains a unique identifier. When a person enters the attendance area (e.g., a classroom or office), they simply need to bring their RFID card within range of an RFID reader. The reader captures the information from the card and automatically registers the person's attendance in the system. This eliminates the need for manual check-ins and ensures accurate, real-time attendance tracking.

The proposed RFID-based attendance system offers several benefits over traditional methods. It reduces human error and minimizes the risk of fraudulent attendance, such as buddy punching (where one person signs in for another). The system also saves valuable time by eliminating the need for manual entry or the collection of signatures. Moreover, it is an efficient solution, capable of processing a large number of people in a short period of time, which is particularly useful in environments with large populations, such as universities, schools, or corporate offices.

The RFID attendance system prototype that has been developed demonstrates the effectiveness of the approach in streamlining attendance management. The design is simple yet robust, incorporating a user-friendly interface that allows easy integration with existing software systems for record-keeping. It is also relatively inexpensive to implement, making it an attractive option for both commercial and academic institutions seeking to modernize their attendance processes.

Another advantage of this system is its portability. The compact size of RFID readers and tags allows the system to be deployed in various locations, whether it's for a small classroom, a large conference room, or across multiple facilities in a larger campus or corporate environment. This flexibility makes RFID technology a versatile tool for enhancing efficiency in a wide range of settings.

Overall, this RFID-based attendance system offers a cost-effective, scalable, and reliable solution for improving the accuracy, speed, and convenience of attendance management. By reducing manual tasks, increasing automation, and providing a seamless experience for both administrators and participants, it stands as a promising solution for the modernized workplace and educational institution.

KEYWORDS :

RFID, ARDUINO, IR SENSOR, LCD, LDR, BUZZER, LED.

LIST OF THE CONTENTS

TITLE	PAGE. NO
ABSTRACT	
CHAPTER 1 : INTRODUCTION	1 - 2
1.1 Relevance of the project	
1.2 Problem Statement	
1.3 Components	
CHAPTER 2 : EXISTED SYSTEMS	3 - 5
CHAPTER 3 : PROPOSED SYSTEMS	5 – 7
CHAPTER 4 : SYSTEM FLOW	
4.1 Block Diagram	8 - 9
CHAPTER 5 : ARDUINO SOFTWARE	9 – 15
5.1 Writing Sketches	10
5.2 Sketchbook	11
5.3 Tabs, Multiple Files	12
5.4 Uploading	13
5.5 Libraries	14
5.6 Third-Party Hardware	15
CHAPTER 6 : ARDUINO HARDWARE	16 – 27
CHAPTER 7 : COMPONENTS	28 – 56
CHAPTER 8 : FEATURES	56 – 58
CHAPTER 9 : ADVANTAGES & DISADVANTAGES	58 – 61
CHAPTER 10 : ARDUINO CODE	62 – 65
CHAPTER 11 : CONCLUSION & REFERENCES	66 - 68

Smart Attend: RFID-Enabled Attendance System for Automated Employee or Student Tracking

1 . INTRODUCTION :

Radio Frequency Identification (RFID) technology has seen widespread adoption across various sectors, including transportation, agriculture, manufacturing, and retail, due to its ability to automate data retrieval without the need for physical contact. This technology is continuously evolving, and its application has become increasingly prevalent in industries seeking efficient and time-saving solutions for routine tasks. One such task that has always been a burden in educational institutions is attendance-taking. Traditionally, attendance is recorded manually, which can be extremely time-consuming, prone to human error, and often inefficient, especially in environments with large numbers of students or employees.

Our project aims to address this issue by utilizing RFID technology to automate and streamline the attendance process. The core concept behind this RFID-based attendance system is to eliminate the manual intervention involved in attendance logging, making the process faster, more accurate, and highly reliable. This system not only benefits educational institutions but can also be adapted for commercial applications where efficient attendance tracking is crucial.

1.1 Relevance of the Project

Radio Frequency Identification (RFID) is a technology that uses electromagnetic fields to automatically identify and track tags attached to objects. These RFID tags contain data that can be read from a distance by an RFID reader. RFID tags can be either active or passive, with passive RFID tags being more commonly used in attendance systems. Passive tags do not have their own power source; they rely on the energy emitted from the RFID reader to transmit data. These tags are inexpensive, compact, and have a relatively short range, making them ideal for applications like attendance systems.

An RFID system consists of several key components:

- RFID Tags – These are small devices that contain unique identification numbers, which are assigned to each individual (in this case, students or employees).
- RFID Reader – The reader emits radio waves that power the RFID tags and captures the unique data from each tag.
- Backend Storage System – The captured data is processed and stored in a central system, which in this project involves storing the information in an SD card file and ultimately sending it to a cloud-based platform for further analysis.
- Middleware/Controller – This part of the system connects all the hardware elements and ensures the smooth communication between the RFID reader, backend system, and other components.

1.2 The Problem with Manual Attendance

In many educational institutions, taking attendance manually is an essential yet time-consuming task. Teachers often spend a significant portion of their class period calling out names, waiting for students to respond, and marking attendance on paper or digital systems. Not only does this waste valuable instructional time, but it also opens the door to errors such as incorrect names being recorded, students skipping attendance, or even fraud, where one student might mark another as present.

Additionally, for large classes or workplaces, managing attendance records manually becomes increasingly difficult to keep track of. Teachers or administrators must often cross-check information from different sources, leading to inefficiency and confusion.

This project aims to resolve this problem by implementing an automated system that ensures a smooth and accurate attendance-taking process. By adopting RFID technology, we can automate the attendance recording process and reduce the burden on teachers and administrators.

System Design: Key Features

The RFID-based attendance system designed in this project involves several steps to ensure the attendance process is streamlined, secure, and efficient. Here's how the system works:

- Unique RFID Tags for Each User: Each individual (such as a student or employee) is assigned a unique RFID tag. This tag could be a physical card, wristband, or keychain with an embedded RFID chip that contains a unique identifier. This tag is then associated with the individual's profile in the backend system, ensuring that their attendance is accurately recorded each time the tag is scanned.
- Automatic Attendance Recording: When the individual enters the designated attendance area (e.g., classroom, office, or other locations), they simply need to bring their RFID tag within range of the RFID reader. The reader will then capture the unique ID stored on the tag, along with the time and date of entry, and register this information in the system.
- Data Storage and Processing: The captured data, including the unique ID of the person, timestamp, and attendance status (present/absent), is stored in an SD card within the system. This ensures that the attendance data is securely recorded even if there is a temporary interruption in internet connectivity.
- Integration with Cloud Storage: At the end of the day or once the session is complete, the system sends the recorded attendance data to a cloud-based platform, in this case, the Thingspeak channel. This platform functions as an API application that collects, stores, and analyzes the data remotely. By using Wi-Fi, the system automatically uploads the attendance data to Thingspeak, making it accessible from any device with internet access.
- User-Friendly Interface: The RFID-based attendance system has been designed with both the user and administrator in mind. The process of checking in with an RFID tag is quick and effortless, reducing the time it takes to record attendance. The backend system, on the other hand, provides an intuitive interface for administrators to view, analyze, and export attendance data as needed.
- Time and Date Tracking: One of the most important features of this system is its ability to accurately track not just the presence of an individual but also the exact time and date when attendance was recorded. This data can be used for reporting, monitoring attendance trends, or managing work hours in commercial environments.

- Wi-Fi Connectivity for Data Upload: Once attendance is recorded, the system uses Wi-Fi connectivity to upload the data to Thingspeak. This cloud-based platform provides a central location for managing attendance records and allows for real-time monitoring. The data can also be visualized using various analytical tools available within Thingspeak, making it easy to generate reports and insights.

1.3 Components

This project involves both hardware and software components working in tandem to achieve the desired objectives. Let's break down these elements:

Hardware Components:

- RFID Reader and Tags: The primary hardware component is the RFID reader, which communicates with the tags and retrieves the unique identifiers. The tags themselves are passive and contain only the identification number of the individual.
- Microcontroller: A microcontroller, typically an Arduino or Raspberry Pi, is used to control the overall system. It processes the data from the RFID reader and manages the storage and upload of attendance records.
- Wi-Fi Module: A Wi-Fi module, such as the ESP8266 or ESP32, is used to enable wireless communication between the system and the Things peak platform, allowing the attendance data to be uploaded in real time.
- SD Card Module: This is used to store the data locally before it is uploaded to the cloud, ensuring no data is lost in case of internet connectivity issues.

Software Components:

- Arduino IDE or Python: The microcontroller is programmed using an integrated development environment (IDE), typically the Arduino IDE or Python, to control the RFID reader and manage the upload process to Things peak.
- Things peak API: The Things peak platform provides a cloud-based API for receiving and storing the uploaded attendance data. It also allows for real-time data analysis and visualization.
- Database Management: Data is organized and stored efficiently using the SD card and Things peak platform, where each record includes unique identifiers, timestamps, and attendance status.

2. EXISTING SYSTEMS

Barcodes: A Foundation of Inventory Management

Barcodes, those ubiquitous patterns of black bars and white spaces, have revolutionized how we track and manage goods. As automated scanning systems, they encode information about a product into a machine-readable format. This information, typically a unique identifier, allows for efficient inventory tracking, pricing, and point-of-sale transactions.

How Barcodes Work

A barcode scanner emits a beam of light that is reflected back at varying intensities depending on the pattern of bars and spaces. This information is then translated into a digital code, which is linked to specific product data in a database. This data can include the product name, manufacturer, price, and other relevant details.

Applications of Barcodes

Barcodes are ubiquitous in our daily lives. They are found on a wide range of products, from groceries and clothing to electronics and pharmaceuticals. They are also used in various industries, including retail, logistics, healthcare, and manufacturing.

Advantages of Barcodes

- Cost-effectiveness: Barcodes are relatively inexpensive to produce and integrate into existing systems.
- Reliability: Barcode scanners are highly accurate and can quickly read large volumes of data.
- Global Recognition: The barcode system is a widely adopted standard, ensuring compatibility across different systems and regions.

Disadvantages of Barcodes

- Limited Data Capacity: Barcodes can only store a limited amount of information, restricting their use in complex applications.
- Vulnerability to Damage: Physical damage to the barcode can render it unreadable.
- Line-of-Sight Requirement: Barcode scanners require a direct line of sight to the barcode, which can be challenging in certain situations.
- Security Concerns: Barcodes can be easily replicated, raising concerns about counterfeiting and fraud.

RFID: A Leap Forward in Tracking Technology

Radio-frequency identification (RFID) technology offers a significant advancement over traditional barcodes. It utilizes radio waves to communicate between a tag attached to an object and a reader. This enables contactless, real-time tracking and identification of objects.

How RFID Works

RFID tags contain a microchip and an antenna. When an RFID reader emits a radio signal, the tag responds by transmitting its unique identification code. This information can then be used to track the object's location, movement, and other relevant data.

Advantages of RFID

- No Line-of-Sight Requirement: RFID readers can read tags even when they are not in direct line of sight, making it ideal for tracking items in cluttered environments.
- High-Speed Reading: RFID systems can read multiple tags simultaneously, significantly increasing efficiency.
- Data Capacity: RFID tags can store much more data than barcodes, enabling more detailed tracking and information storage.
- Durability: RFID tags are generally more durable than barcodes and can withstand harsh environments.
- Enhanced Security: RFID tags can be encrypted, making them more secure against counterfeiting and tampering.

Applications of RFID

RFID technology is finding increasing applications in various sectors, including:

- Supply Chain Management: Tracking goods throughout the supply chain, from manufacturing to delivery.
- Retail: Inventory management, loss prevention, and self-checkout systems.
- Healthcare: Patient identification, medication tracking, and asset management.
- Transportation: Toll collection, vehicle tracking, and access control.
- Logistics: Real-time tracking of shipments and containers.

Choosing Between RFID and Barcodes

The choice between RFID and barcodes depends on the specific application requirements.

- Cost: Barcodes are generally more cost-effective for simple applications with low data requirements.
- Data Capacity: RFID is preferred when extensive data storage and complex tracking are needed.
- Durability: RFID is more suitable for applications involving harsh environments or frequent handling.
- Security: RFID offers enhanced security features compared to barcodes.
- Read Range: RFID provides greater flexibility in reading distances, making it ideal for applications where line-of-sight is limited.

3. PROPOSED SYSTEMS :

System Architecture:

1. Microcontroller: The Arduino Uno is a popular choice due to its ease of use, availability, and extensive libraries. However, other microcontrollers like ESP32 can also be used, offering Wi-Fi capabilities for cloud connectivity.
2. RFID Module: The EM-18 is a commonly used RFID module, but other options like RC522 are also suitable. The choice depends on factors like read range, communication protocol, and power consumption.
3. Sensors:
    1. IR Sensor: A passive infrared (PIR) sensor is used to detect motion and ensure that an actual person is present in front of the reader. This prevents accidental or false readings.
    2. Flame Sensor: A flame sensor (such as a KY-028) detects infrared radiation emitted by flames. This adds a crucial safety layer, especially in environments where fire hazards are a concern.
4. User Interface:
    1. LCD Display: A 16x2 character LCD provides clear and concise feedback messages to the user, such as "Attendance Registered," "Invalid Tag," "System Initialized," or "Fire Alert."

    2. LEDs: Green and Red LEDs provide visual cues for successful and failed attendance attempts, respectively.
    3. Buzzer: An active buzzer provides audible feedback, further enhancing the user experience and alerting users to system status.
5. Data Storage:
    1. Internal Memory: The Arduino's internal memory can store a limited number of user IDs and attendance records.
    2. SD Card: An SD card can be used for larger storage capacity, allowing for long-term data logging.
    3. External Database: For more robust data management and analysis, an external database like MySQL or SQLite can be integrated. Cloud databases (e.g., Firebase, AWS) can be used for remote data access and real-time monitoring.

System Operation:

6. Power-On and Initialization:
    1. Upon powering up, the Arduino initializes all components: RFID reader, IR sensor, flame sensor, LCD, LEDs, and buzzer.
    2. It loads the user database (if using an external database) or checks for pre-stored user IDs in its memory.
    3. The system displays an initialization message on the LCD.
7. User Presence Detection:
    1. The IR sensor monitors for motion in front of the RFID reader.
    2. If no motion is detected, the system waits for user presence.
    3. Once motion is detected, the system is ready to scan the RFID tag.
8. RFID Tag Scanning:
    1. When a user presents their RFID tag to the reader, the module reads the unique ID (UID).
    2. The Arduino receives the UID and compares it with the stored user database.
9. Attendance Verification:
    1. Valid Tag:
        1. If the UID matches a valid user record, the system logs the attendance time.
        2. The Green LED turns ON.
        3. "Attendance Registered" is displayed on the LCD.
        4. The buzzer emits a short beep.
        5. Attendance data is logged (internally or to an external database).
    2. Invalid Tag:
        1. If the UID is invalid or not found in the database:

        1. The Red LED turns ON.
        2. "Invalid Tag" or "Access Denied" is displayed on the LCD.
        3. The buzzer emits a longer, warning beep.

10. Safety Monitoring:
    1. The flame sensor continuously monitors the environment for fire hazards.
    2. If a flame is detected:
        1. The Red LED blinks rapidly.
        2. The buzzer emits a continuous, loud alarm.
        3. An alert message ("Fire Detected!") is displayed on the LCD.
        4. The system can be programmed to shut down or activate a fire suppression mechanism (if applicable).
11. Data Management:
    1. Attendance data can be stored in various ways:
        1. Internally in the Arduino's memory (for limited storage).
        2. On an SD card for larger storage capacity.
        3. In an external database (MySQL, SQLite, cloud databases) for centralized data management, analysis, and reporting.
12. User Interface:
    1. The LCD provides real-time feedback to the user, informing them of the system's status, attendance success/failure, and any alerts.
    2. LEDs provide visual cues for system status and attendance outcomes.
    3. The buzzer provides audible alerts for successful attendance, failed attempts, and safety warnings.

Additional Considerations:

- Security:
    1. Implement robust security measures to prevent unauthorized access to the system and data.
    2. Consider using encryption for sensitive data.
- Network Connectivity:
    1. If using an ESP32 or other Wi-Fi-enabled microcontroller, integrate with a local network or the cloud for remote monitoring, data

4. SYSTEM FLOWCHART :

Start

↓

Power On → Initialize RFID Reader, Sensors, LCD, LEDs, and Buzzer

↓

Check IR Sensor (Presence Detection)

↓

No Presence → Wait for User

↓

Presence Detected → Scan RFID Tag → Get UID

↓

Verify UID →

↓ ↓

Valid UID → Log Attendance → Green LED, LCD ("Attendance Registered"), Buzzer (short beep)

↓

Invalid UID → Display "Invalid Tag" on LCD → Red LED, Buzzer (long beep)

↓

Monitor Flame Sensor for Safety

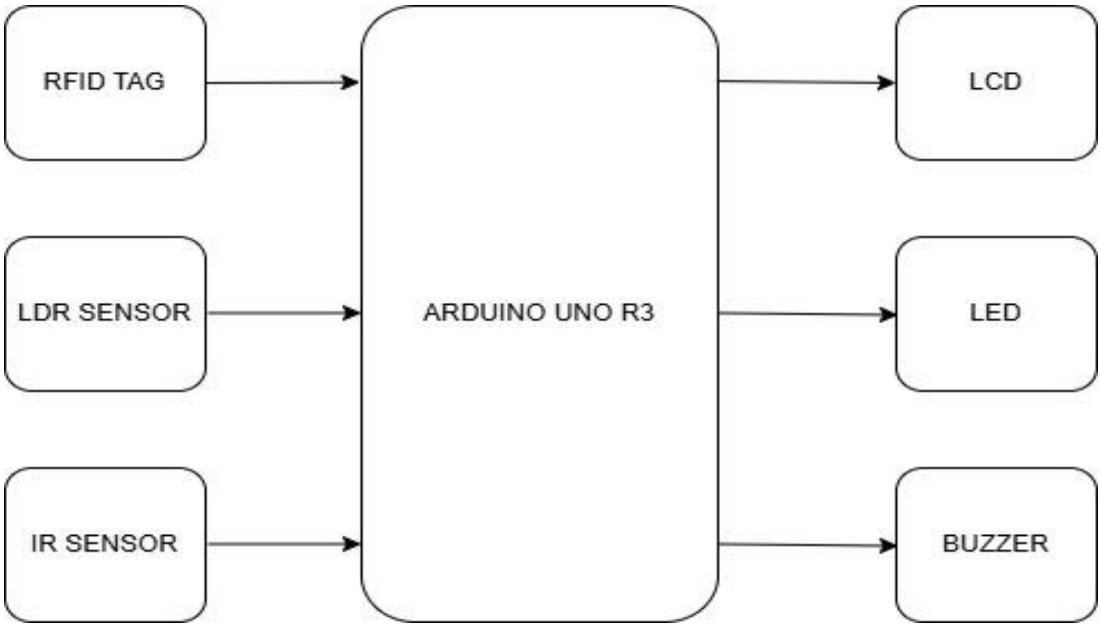
↓

Flame Detected → Trigger Alarm (Red LED blink, Buzzer continuous beep)

↓

End

4.1 BLOCK DIAGRAM :



5. ARDUINO SOFTWARE :

Arduino Software (IDE)

1. Writing Sketches
2. File
3. Edit
4. Sketch
5. Tools
6. Help
7. Sketchbook
8. Tabs, Multiple Files, and Compilation
9. Uploading
10. Libraries
11. Third-Party Hardware
12. Serial Monitor
13. Preferences
14. Language Support
15. Boards

The Arduino Integrated Development Environment - or Arduino Software (IDE) - contains a text editor for writing code, a message area, a text console, a toolbar with buttons for common functions and a series of menus. It connects to the Arduino and Genuino hardware to upload programs and communicate with them.

5.1 Writing Sketches

Programs written using Arduino Software (IDE) are called sketches. These sketches are written in the text editor and are saved with the file extension .ino. The editor has features for cutting/pasting and for searching/replacing text. The message area gives feedback while saving and exporting and also displays errors. The console displays text output by the Arduino Software (IDE), including complete error messages and other information. The bottom righthand corner of the window displays the configured board and serial port. The toolbar buttons allow you to verify and upload programs, create, open, and save sketches, and open the serial monitor.

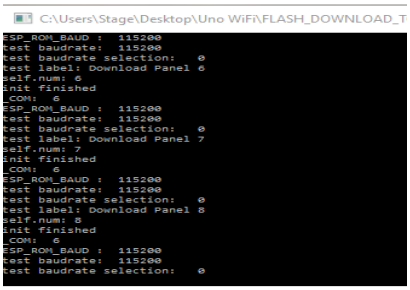
NB: Versions of the Arduino Software (IDE) prior to 1.0 saved sketches with the extension .pde. It is possible to open these files with version 1.0, you will be prompted to save the sketch with the.ino extension on save.

Verify
Checks your code for errors compiling it.

Upload
Compiles your code and uploads it to the configured board. See uploading below for details.

Note: If you are using an external programmer with your board, you can hold down the "shift" key on your computer when using this icon. The text will change to "Upload using Programmer"

New
Creates a new sketch.



Open
Presents a menu of all the sketches in your sketchbook. Clicking one will open it within the current window overwriting its content.

Note: due to a bug in Java, this menu doesn't scroll; if you need to open a sketch late in the list, use the File | Sketchbookmenu instead.

Save
Saves your sketch.

Serial Monitor
Opens the serial monitor.

Additional commands are found within the five menus: File, Edit, Sketch, Tools, Help. The menus are context sensitive, which means only those items relevant to the work currently being carried out are available.

File

1. New
   Creates a new instance of the editor, with the bare minimum structure of a sketch already in place.
2. Open
   Allows to load a sketch file browsing through the computer drives and folders.
3. Open Recent
   Provides a short list of the most recent sketches, ready to be opened.
4. Sketchbook
   Shows the current sketches within the sketchbook folder structure; clicking on any name opens the corresponding sketch in a new editor instance.
5. Examples
   Any example provided by the Arduino Software (IDE) or library shows up in this menu item. All the examples are structured in a tree that allows easy access by topic or library.
6. Close
   Closes the instance of the Arduino Software from which it is clicked.
7. Save
   Saves the sketch with the current name. If the file hasn't been named before, a name will be provided in a "Save as.." window.
8. Save as...
   Allows to save the current sketch with a different name.
9. Page Setup
   It shows the Page Setup window for printing.
10. Print
    Sends the current sketch to the printer according to the settings defined in Page Setup.
11. Preferences
    Opens the Preferences window where some settings of the IDE may be customized, as the language of the IDE interface.
12. Quit
    Closes all IDE windows. The same sketches open when Quit was chosen will be automatically reopened the next time you start the IDE.

Edit

1. Undo/Redo
   Goes back of one or more steps you did while editing; when you go back, you may go forward with Redo.
2. Cut
   Removes the selected text from the editor and places it into the clipboard.
3. Copy
   Duplicates the selected text in the editor and places it into the clipboard.

4. Copy for Forum
   Copies the code of your sketch to the clipboard in a form suitable for posting to the forum, complete with syntax coloring.
5. Copy as HTML
   Copies the code of your sketch to the clipboard as HTML, suitable for embedding in web pages.
6. Paste
   Puts the contents of the clipboard at the cursor position, in the editor.
7. Select All
   Selects and highlights the whole content of the editor.
8. Comment/Uncomment
   Puts or removes the // comment marker at the beginning of each selected line.
9. Increase/Decrease Indent
   Adds or subtracts a space at the beginning of each selected line, moving the text one space on the right or eliminating a space at the beginning.
10. Find
    Opens the Find and Replace window where you can specify text to search inside the current sketch according to several options.
11. Find Next
    Highlights the next occurrence - if any - of the string specified as the search item in the Find window, relative to the cursor position.
12. Find Previous
    Highlights the previous occurrence - if any - of the string specified as the search item in the Find window relative to the cursor position.

Sketch

1. Verify/Compile
   Checks your sketch for errors compiling it; it will report memory usage for code and variables in the console area.
2. Upload
   Compiles and loads the binary file onto the configured board through the configured Port.
3. Upload Using Programmer
   This will overwrite the bootloader on the board; you will need to use Tools > Burn Bootloader to restore it and be able to Upload to USB serial port again. However, it allows you to use the full capacity of the Flash memory for your sketch. Please note that this command will NOT burn the fuses. To do so a Tools -> Burn Bootloader command must be executed.
4. Export Compiled Binary
   Saves a .hex file that may be kept as archive or sent to the board using other tools.
5. Show Sketch Folder
   Opens the current sketch folder.
6. Include Library
   Adds a library to your sketch by inserting #include statements at the start of your code. For more details, see libraries below. Additionally, from this menu item you can access the Library Manager and import new libraries from .zip files.
7. Add File...
   Adds a source file to the sketch (it will be copied from its current location). The new file appears in a new tab in the sketch window. Files can be removed from the sketch using the tab menu accessible clicking on the small triangle icon below the serial monitor one on the right side o the toolbar.

Tools

1. Auto Format
   This formats your code nicely: i.e. indents it so that opening and closing curly braces line up, and that the statements inside curly braces are indented more.
2. Archive Sketch
   Archives a copy of the current sketch in .zip format. The archive is placed in the same directory as the sketch.
3. Fix Encoding & Reload
   Fixes possible discrepancies between the editor char map encoding and other operating systems char maps.
4. Serial Monitor
   Opens the serial monitor window and initiates the exchange of data with any connected board on the currently selected Port. This usually resets the board, if the board supports Reset over serial port opening.
5. Board
   Select the board that you're using. See below for descriptions of the various boards.
6. Port
   This menu contains all the serial devices (real or virtual) on your machine. It should automatically refresh every time you open the top-level tools menu.
7. Programmer
   For selecting a harware programmer when programming a board or chip and not using the onboard USB-serial connection. Normally you won't need this, but if you're burning a bootloader to a new microcontroller, you will use this.
8. Burn Bootloader
   The items in this menu allow you to burn a bootloader onto the microcontroller on an Arduino board. This is not required for normal use of an Arduino or Genuino board but is useful if you purchase a new ATmega microcontroller (which normally come without a bootloader). Ensure that you've selected the correct board from the Boards menu before burning the bootloader on the target board. This command also set the right fuses.

Help

Here you find easy access to a number of documents that come with the Arduino Software (IDE). You have access to Getting Started, Reference, this guide to the IDE and other documents locally, without an internet connection. The documents are a local copy of the online ones and may link back to our online website.

- Find in Reference This is the only interactive function of the Help menu: it directly selects the relevant page in the local copy of the Reference for the function or command under the cursor.

5.2 Sketchbook

The Arduino Software (IDE) uses the concept of a sketchbook: a standard place to store your programs (or sketches). The sketches in your sketchbook can be opened from the File > Sketchbook menu or from the Open button on the toolbar. The first time you run the Arduino software, it will automatically create a directory for your sketchbook. You can view or change the location of the sketchbook location from with the Preferences dialog.

Beginning with version 1.0, files are saved with a .ino file extension. Previous versions use the .pde extension. You may still open .pde named files in version 1.0 and later, the software will automatically rename the extension to .ino.

5.3 Tabs, Multiple Files, and Compilation

Allows you to manage sketches with more than one file (each of which appears in its own tab). These can be normal Arduino code files (no visible extension), C files (.c extension), C++ files (.cpp), or header files (.h).

5.4 Uploading

Before uploading your sketch, you need to select the correct items from the Tools > Board and Tools > Port menus. The boards are described below. On the Mac, the serial port is probably something like /dev/tty.usbmodem241 (for an Uno or Mega2560 or Leonardo) or /dev/tty.usbserial-1B1 (for a Duemilanove or earlier USB board), or /dev/tty.USA19QW1b1P1.1 (for a serial board connected with a Keyspan USB-to-Serial adapter). On Windows, it's probably COM1 or COM2 (for a serial board) or COM4, COM5, COM7, or higher (for a USB board) - to find out, you look for USB serial device in the ports section of the Windows Device Manager. On Linux, it should be /dev/ttyACMx , /dev/ttyUSBx or similar. Once you've selected the correct serial port and board, press the upload button in the toolbar or select the Upload item from the Sketch menu. Current Arduino boards will reset automatically and begin the upload. With older boards (pre-Diecimila) that lack auto-reset, you'll need to press the reset button on the board just before starting the upload. On most boards, you'll see the RX and TX LEDs blink as the sketch is uploaded. The Arduino Software (IDE) will display a message when the upload is complete, or show an error.

When you upload a sketch, you're using the Arduino bootloader, a small program that has been loaded on to the microcontroller on your board. It allows you to upload code without using any additional hardware. The bootloader is active for a few seconds when the board resets; then it starts whichever sketch was most recently uploaded to the microcontroller. The bootloader will blink the on-board (pin 13) LED when it starts (i.e. when the board resets).

5.5 Libraries

Libraries provide extra functionality for use in sketches, e.g. working with hardware or manipulating data. To use a library in a sketch, select it from the Sketch > Import Library menu. This will insert one or more #include statements at the top of the sketch and compile the library with your sketch. Because libraries are uploaded to the board with your sketch, they increase the amount of space it takes up. If a sketch no longer needs a library, simply delete its #includestatements from the top of your code.

There is a list of libraries in the reference. Some libraries are included with the Arduino software. Others can be downloaded from a variety of sources or through the Library Manager. Starting with version 1.0.5 of the IDE, you do can import a library from a zip file and use it in an open sketch. See these instructions for installing a third-party library.

To write your own library, see this tutorial.

5.6 Third-Party Hardware

Support for third-party hardware can be added to the hardware directory of your sketchbook directory. Platforms installed there may include board definitions (which appear in the board menu), core libraries, bootloaders, and programmer definitions. To install, create the hardware directory, then unzip the third-party platform into its own sub-directory. (Don't use "arduino" as the sub-directory name or you'll override the built-in Arduino platform.) To uninstall, simply delete its directory.

For details on creating packages for third-party hardware, see the Arduino IDE 1.5 3rd party Hardware specification.

Serial Monitor

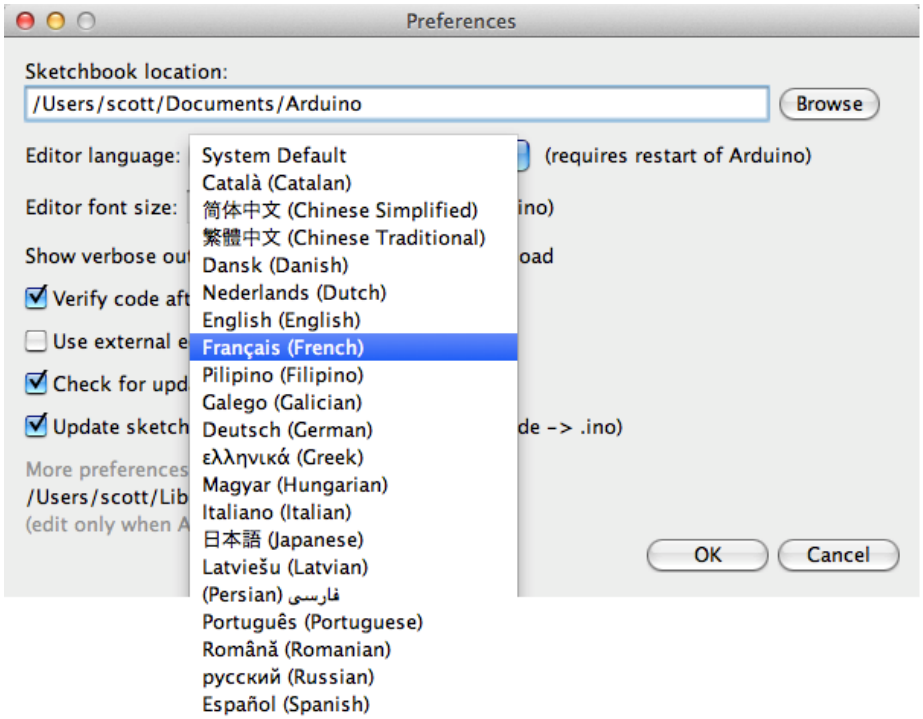
Displays serial data being sent from the Arduino or Genuino board (USB or serial board). To send data to the board, enter text and click on the "send" button or press enter. Choose the baud rate from the drop-down that matches the rate passed to Serial.begin in your sketch. Note that on Windows, Mac or Linux, the Arduino or Genuino board will reset (rerun your sketch execution to the beginning) when you connect with the serial monitor.

You can also talk to the board from Processing, Flash, MaxMSP, etc (see the interfacing page for details).

Preferences

Some preferences can be set in the preferences dialog (found under the Arduino menu on the Mac, or File on Windows and Linux). The rest can be found in the preferences file, whose location is shown in the preference dialog.

Language Support



Since version 1.0.1 , the Arduino Software (IDE) has been translated into 30+ different languages. By default, the IDE loads in the language selected by your operating system. (Note: on Windows and possibly Linux, this is determined by the locale setting which controls currency and date formats, not by the language the operating system is displayed in.)

If you would like to change the language manually, start the Arduino Software (IDE) and open the Preferences window. Next to the Editor Language there is a dropdown menu of currently supported languages. Select your preferred language from the menu, and restart the software to use the selected language. If your operating system language is not supported, the Arduino Software (IDE) will default to English.

You can return the software to its default setting of selecting its language based on your operating system by selecting System Default from the Editor Language drop-down. This setting will take effect when you restart the Arduino Software (IDE). Similarly, after changing your operating system's settings, you must restart the Arduino Software (IDE) to update it to the new default language.

Boards

The board selection has two effects: it sets the parameters (e.g. CPU speed and baud rate) used when compiling and uploading sketches; and sets and the file and fuse settings used by the burn bootloader command. Some of the board definitions differ only in the latter, so even if you've been uploading successfully with a particular selection you'll want to check it before burning the bootloader. You can find a comparison table between the various boards here.

Arduino Software (IDE) includes the built in support for the boards in the following list, all based on the AVR Core. The Boards Manager included in the standard installation allows to add support for the growing number of new boards based on different cores like Arduino Due, Arduino Zero, Edison, Galileo and so on.

1. Arduino Yùn An ATmega32u4 running at 16 MHz with auto-reset, 12 Analog In, 20 Digital I/O and 7 PWM.
2. Arduino/Genuino Uno An ATmega328 running at 16 MHz with auto-reset, 6 Analog In, 14 Digital I/O and 6 PWM.
3. Arduino Diecimila or Duemilanove w/ ATmega168An ATmega168 running at 16 MHz with auto-reset.
4. Arduino Nano w/ ATmega328 An ATmega328 running at 16 MHz with auto-reset. Has eight analog inputs.
5. Arduino/Genuino Mega 2560 An ATmega2560 running at 16 MHz with auto-reset, 16 Analog In, 54 Digital I/O and 15 PWM.
6. Arduino Mega An ATmega1280 running at 16 MHz with auto-reset, 16 Analog In, 54 Digital I/O and 15 PWM.
7. Arduino Mega ADK An ATmega2560 running at 16 MHz with auto-reset, 16 Analog In, 54 Digital I/O and 15 PWM.
8. Arduino Leonardo An ATmega32u4 running at 16 MHz with auto-reset, 12 Analog In, 20 Digital I/O and 7 PWM.
9. Arduino/Genuino Micro An ATmega32u4 running at 16 MHz with auto-reset, 12 Analog In, 20 Digital I/O and 7 PWM.
10. Arduino Esplora An ATmega32u4 running at 16 MHz with auto-reset.
11. Arduino Mini w/ ATmega328An ATmega328 running at 16 MHz with auto-reset, 8 Analog In, 14 Digital I/O and 6 PWM.
12. Arduino Ethernet Equivalent to Arduino UNO with an Ethernet shield: An ATmega328 running at 16 MHz with auto-reset, 6 Analog In, 14 Digital I/O and 6 PWM.
13. Arduino Fio An ATmega328 running at 8 MHz with auto-reset. Equivalent to Arduino Pro or Pro Mini (3.3V, 8 MHz) w/ ATmega328, 6 Analog In, 14 Digital I/O and 6 PWM.
14. Arduino BT w/ ATmega328 ATmega328 running at 16 MHz. The bootloader burned (4 KB) includes codes to initialize the on-board bluetooth module, 6 Analog In, 14 Digital I/O and 6 PWM..
15. LilyPad Arduino USB An ATmega32u4 running at 8 MHz with auto-reset, 4 Analog In, 9 Digital I/O and 4 PWM.
16. LilyPad Arduino An ATmega168 or ATmega132 running at 8 MHz with auto-reset, 6 Analog In, 14 Digital I/O and 6 PWM.
17. Arduino Pro or Pro Mini (5V, 16 MHz) w/ ATmega328 An ATmega328 running at 16 MHz with auto-reset. Equivalent to Arduino Duemilanove or Nano w/ ATmega328; 6 Analog In, 14 Digital I/O and 6 PWM.

18. ArduinoNG or older w/ ATmega168 An ATmega168 running at 16 MHz without auto-reset. Compilation and upload is equivalent to Arduino Diecimila or Duemilanove w/ ATmega168, but the bootloader burned has a slower timeout (and blinks the pin 13 LED three times on reset); 6 Analog In, 14 Digital I/O and 6 PWM.

19. Arduino Robot Control An ATmega328 running at 16 MHz with auto-reset.
20. Arduino Robot Motor An ATmega328 running at 16 MHz with auto-reset.
21. Arduino Gemma An ATtiny85 running at 8 MHz with auto-reset, 1 Analog In, 3 Digital I/O and 2 PWM.

6. ARDUINO HARDWARE :

Arduino

The Arduino Uno is an open-source microcontroller board based on the Microchip ATmega328P microcontroller and developed by Arduino.cc. The board is equipped with sets of digital and analog input/output (I/O) pins that may be interfaced to various expansion boards (shields) and other circuits. The board has 14 Digital pins, 6 Analog pins, and programmable with the Arduino IDE (Integrated Development Environment) via a type B USB cable. It can be powered by the USB cable or by an external 9-volt battery, though it accepts voltages between 7 and 20 volts. It is also similar to the Arduino Nano and Leonardo. You can tell your board what to do by sending a set of instructions to the microcontroller on the board. To do so you use the Arduino programming language (based on Wiring), and the Arduino Software (IDE), based on Processing.

ARDUINO:

setup( ): A function present in every Arduino sketch. Run once before the loop( ) function. Often used to set pinmode to input or output. The setup( ) function looks like:

```
void setup( ){
        //code goes here
        }
```

loop( ): A function present in every single Arduino sketch. This code happens over and over again. The loop( ) is where (almost) everything happens. The one exception to this is setup( ) and variable declaration. ModKit uses another type of loop called “forever( )” which executes over Serial. The loop( ) function looks like:

```
void loop( ) {
        //code goes here
        }
```

input: A pin mode that intakes information.

output: A pin mode that sends information.

HIGH: Electrical signal present (5V for Uno). Also ON or True in boolean logic.

LOW: No electrical signal present (0V). Also OFF or False in boolean logic.

digitalRead: Get a HIGH or LOW reading from a pin already declared as an input.

digitalWrite: Assign a HIGH or LOW value to a pin already declared as an output.

analogRead: Get a value between or including 0 (LOW) and 1023 (HIGH). This allows you to get readings from analog sensors or interfaces that have more than two states.

analogWrite: Assign a value between or including 0 (LOW) and 255 (HIGH). This allows you to set output to a PWM value instead of just HIGH or LOW.

PWM: Stands for Pulse-Width Modulation, a method of emulating an analog signal through a digital pin. A value between or including 0 and 255. Used with analogWrite.

These boards below use the same micro-controller, just in a different package. The Lilypad is designed for use with conductive thread instead of wire and the Arduino Mini is simply a smaller package without the USB, Barrel Jack and Power Outs.

It depends on what you want to do with it really. There are two different purposes outlined above for the voltage divider, we will go over both.

If you wish to use the voltage divider as a sensor reading device first you need to know the maximum voltage allowed by the analog inputs you are using to read the signal. On an Arduino this is 5V. So, already we know the maximum value we need for Vout. The Vin is simply the amount of voltage already present on the circuit before it reaches the first resistor. You should be able to find the maximum voltage your sensor outputs by looking on the Datasheet; this is the maximum amount of voltage your sensor will let through given the voltage in of your circuit. Now we have exactly one variable left, the value of the second resistor. Solve for R2 and you will have all the components of your voltage divider figured out! We solve for R1's highest value because a smaller resistor will simply give us a smaller signal which will be readable by our analog inputs.

Powering an analog Reference is exactly the same as reading a sensor except you have to calculate for the Voltage Out value you want to use as the analog Reference.

All of the electrical signals that the Arduino works with are either Analog or Digital. It is extremely important to understand the difference between these two types of signal and how to manipulate the information these signals represent.

DIGITAL

An electronic signal transmitted as binary code that can be either the presence or absence of current, high and low voltages or short pulses at a particular frequency.

Humans perceive the world in analog, but robots, computers and circuits use Digital. A digital signal is a signal that has only two states. These states can vary depending on the signal, but simply defined the states are ON or OFF, never in between.

In the world of Arduino, Digital signals are used for everything with the exception of Analog Input. Depending on the voltage of the Arduino the ON or HIGH of the Digital signal will be equal to the system voltage, while the OFF or LOW signal will always equal 0V. This is a fancy way of saying that on a 5V Arduino the HIGH signals will be a little under 5V and on a 3.3V Arduino the HIGH signals will be a little under 3.3V.

To receive or send Digital signals the Arduino uses Digital pins # 0 - # 13. You may also setup your Analog In pins to act as Digital pins. To set up Analog In pins as Digital pins use the command:

1. pinMode(pinNumber, value);

Where pinNumber is an Analog pin (A0 – A5) and value is either INPUT or OUTPUT. To setup Digital pins use the same command but reference a Digital pin for pinNumber instead of an Analog In pin. Digital pins default as input, so really you only need to set them to OUTPUT in pinMode. To read these pins use the command:

2. digitalRead(pinNumber);

Where pinNumber is the Digital pin to which the Digital component is connected. The digitalRead command will return either a HIGH or a LOW signal. To send a Digital signal to a pin uses the command:

3. digitalWrite(pinNumber, value);

Where pinNumber is the number of the pin sending the signal and value is either HIGH or LOW.

The Arduino also has the capability to output a Digital signal that acts as an Analog signal; this signal is called Pulse Width Modulation (PWM). Digital Pins # 3, # 5, # 6, # 9, # 10 and #11 have PWM capabilities. To output a PWM signal use the command:

4. analogWrite(pinNumber, value);

Where pinNumber is a Digital Pin with PWM capabilities and value is a number between 0 (0%) and 255 (100%). For more information on PWM see the PWM worksheets or S.I.K. circuit 12.

THINGS TO REMEMBER ABOUT DIGITAL:

1. Digital Input/Output uses the Digital pins, but Analog In pins can be used as Digital
2. To receive a Digital signal use: digitalRead(pinNumber);

3. To send a Digital signal use: digitalWrite(pinNumber, value);
4. Digital Input and Output are always either HIGH or LOW

All of the electrical signals that the Arduino works with are either Analog or Digital. It is extremely important to understand the difference between these two types of signal and how to manipulate the information these signals represent.

ANALOG

Humans perceive the world in analog. Everything we see and hear is a continuous transmission of information to our senses. The temperatures we perceive are never 100% hot or 100% cold, they are constantly changing between our ranges of acceptable temperatures. (And if they are out of our range of acceptable temperatures then what are we doing there?) This continuous stream is what defines analog data. Digital information, the complementary concept to Analog, estimates analog data using only ones and zeros.

In the world of Arduino an Analog signal is simply a signal that can be HIGH (on), LOW (off) or anything in between these two states. This means an Analog signal has a voltage value that can be anything between 0V and 5V (unless you mess with the Analog Reference pin). Analog allows you to send output or receive input about devices that run at percentages as well as on and off. The Arduino does this by sampling the voltage signal sent to these pins and comparing it to a voltage reference signal (5V). Depending on the voltage of the Analog signal when compared to the Analog Reference signal the Arduino then assigns a numerical value to the signal somewhere between 0 (0%) and 1023 (100%). The digital system of the Arduino can then use this number in calculations and sketches.

To receive Analog Input the Arduino uses Analog pins # 0 - # 5. These pins are designed for use with components that output Analog information and can be used for Analog Input. There is no setup necessary, and to read them use the command:

5. analogRead(pinNumber);

Where pinNumber is the Analog In pin to which the the Analog component is connected. The analogRead command will return a number including or between 0 and 1023.

The Arduino also has the capability to output a digital signal that acts as an Analog signal; this signal is called Pulse Width Modulation (PWM). Digital Pins # 3, # 5, # 6, # 9, # 10 and #11 have PWM capabilities. To output a PWM signal use the command:

6. analogWrite(pinNumber, value);

Where pinNumber is a Digital Pin with PWM capabilities and value is a number between 0 (0%) and 255 (100%). On the Arduino UNO PWM pins are signified by a ~ sign. For more information on PWM see the PWM worksheets or S.I.K. circuit 12.

THINGS TO REMEMBER ABOUT ANALOG:

1. Analog Input uses the Analog In pins, Analog Output uses the PWM pins
2. To receive an Analog signal use: analogRead(pinNumber);
3. To send a PWM signal use: analogWrite(pinNumber, value);
4. Analog Input values range from 0 to 1023 (1024 values because it uses 10 bits, 210)
5. PWM Output values range from 0 to 255 (256 values because it uses 8 bits, 28)

All of the electrical signals that the Arduino works with are either input or output. It is extremely important to understand the difference between these two types of signal and how to manipulate the information these signals represent.

OUTPUT SIGNALS

Output to the Arduino pins is always Digital, however there are two different types of Digital Output; regular Digital Output and Pulse Width Modulation Output (PWM). Output is only possible with Digital pins # 0 - # 13. The Digital pins are preset as Output pins, so unless the pin was used as an Input in the same sketch, there is no reason to use the pinMode command to set the pin as an Output. Should a situation arise where it is necessary to reset a Digital pin to Output from Input use the command:

1. pinMode(pinNumber, OUTPUT);

Where pinNumber is the Digital pin number set as Output. To send a Digital Output signal use the command:

2. digitalWrite(pinNumber, value);

Where pinNumber is the Digital pin that is outputting the signal and value is the signal. When outputting a Digital signal value can be either HIGH (On) or LOW (Off).

3. analogWrite(pinNumber, value);

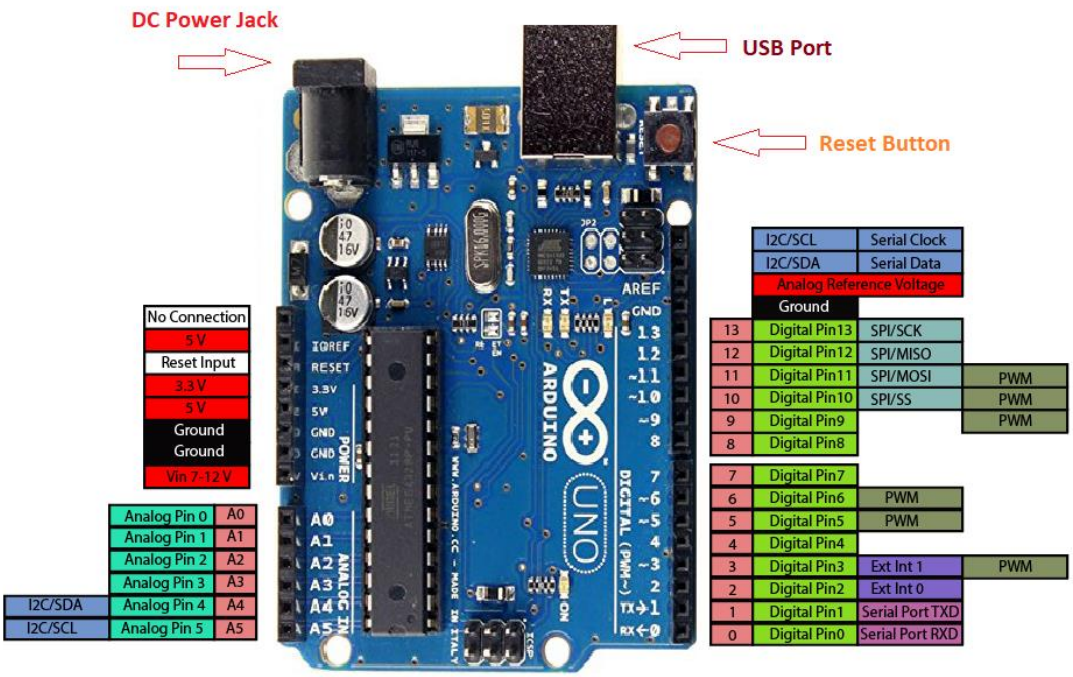
Where pinNumber is a Digital Pin with PWM capabilities and value is a number between 0 (0%) and 255 (100%). For more information on PWM see the PWM worksheets or S.I.K. circuit 12.

Output can be sent to many different devices, but it is up to the user to figure out which kind of Output signal is needed, hook up the hardware and then type the correct code to properly use these signals.

THINGS TO REMEMBER ABOUT OUTPUT:

- Output is always Digital
- There are two kinds of Output: regular Digital or PWM (Pulse Width Modulation)
- To send an Output signal use analogWrite(pinNumber, value); (for analog) or digitalWrite(pinNumber, value); (for digital)
- Output pin mode is set using the pinMode command: pinMode(pinNumber, OUTPUT);
- Regular Digital Output is always either HIGH or LOW
- PWM Output varies from 0 to 255

All of the electrical signals that the Arduino works with are either input or output. It is extremely important to understand the difference between these two types of signal and how to manipulate the information these signals represent.



INPUT SIGNALS

Analog Input enters your Arduino through the Analog In pins # 0 - # 5. These signals originate from analog sensors and interface devices. These analog sensors and devices use voltage levels to communicate their information instead of a simple yes (HIGH) or no (LOW). For this reason you cannot use a digital pin as an input pin for these devices. Analog Input pins are used only for receiving Analog signals. It is only possible to read the Analog Input pins so there is no

command necessary in the setup( ) function to prepare these pins for input. To read the Analog Input pins use the command:

AnalogRead (pinNumber);

Where pinNumber is the Analog Input pin number. This function will return an Analog Input reading between 0 and 1023. A reading of zero corresponds to 0 Volts and a reading of 1023 corresponds to 5 Volts. These voltage values are emitted by the analog sensors and interfaces. If you have an Analog Input that could exceed Vcc + .5V you may change the voltage that 1023 corresponds to by using the Aref pin. This pin sets the maximum voltage parameter your Analog Input pins can read. The Aref pin's preset value is 5V.

Digital Input can enter your Arduino through any of the Digital Pins # 0 - # 13. Digital Input signals are either HIGH (On, 5V) or LOW (Off, 0V). Because the Digital pins can be used either as input or output you will need to prepare the Arduino to use these pins as inputs in your

setup( )function. To do this types the command:

pinMode(pinNumber, INPUT);

inside the curly brackets of the setup( ) function where pinNumber is the Digital pin number you wish to declare as an input. You can change the pinMode in the loop( )function if you need to switch a pin back and forth between input and output, but it is usually set in the setup( )function and left untouched in the loop( )function. To read the Digital pins set as inputs use the command:

digitalRead(pinNumber);

Where pinNumber is the Digital Input pin number.

Input can come from many different devices, but each device's signal will be either Analog or Digital, it is up to the user to figure out which kind of input is needed, hook up the hardware and then type the correct code to properly use these signals.

X1:

- DE-9 serial connector

Used to connect computer (or other devices) using RS-232 standard. Needs a serial cable, with at least 4 pins connected: 2, 3, 4 and 5. Works only when JP0 is set to 2-3 position.

DC1:

- 2.1 mm. power jack

Used to connect external power source. Centre positive. Voltage Regulator Works with regulated +7 to +20 volts DC (9v. to 12v. is recommended). It is possible to alternatively connect external power using 9v. pin or 5v. pin. (see POWER PINOUT)

ICSP:

2x3 pin header Used to program Atmega with bootloader. The number 1 on both sides of the board indicates cable pin1 position. Used to upload sketches on Atmega ICs without bootloader (available only in Arduino IDE versions 0011 and 0012).

JP0 3 pins jumper When in position 2-3, this jumper enables serial connection (through X1 connector) to/from computer/devices. Use this as default position. When in position 1-2, it disables serial communication, and enables external pull-down resistors on pin0 (RX) and pin1 (TX). Use this only to prevent noise on RX (that seems incoming data to Atmega), that sometimes makes sketch not starting. When removing this jumper, serial communication is disabled, and

pin0 and pin1 work as a normal (floating) digital pin. Useful when more digital pins are needed, but only when serial communication is not necessary. External pull-down/pull-up resistor is required.

JP4

2 pins jumper When in position 1-2, this jumper enables auto reset feature, useful when uploading a sketch to Arduino, resetting Atmega automatically. It makes unnecessary to press reset button (S1) when uploading sketches. Be sure that computer COM Port speed is set to 19200bps otherwise auto reset will not work properly. If removed, disables auto reset feature. Very useful to prevent undesired Atmega reset when using sketches that needs serial communication. Auto reset works with DTR pulse on serial pin4. Sometimes Arduino senses a DTR pulse when connecting X1 (serial connector) and some softwares sends a DTR pulse when it starts or when it closes, that makes Atmega reset when not desired.

S1

Tactile button This button resets Atmega, to restart uploaded sketch or to prepare Arduino to receive a sketch through serial connector (when auto reset is not active).

LEDS

Indicative leds POWER led Turns on when Arduino is powered through DC1, +9v. pin or +5v. pin. RX led Blinks when receiving data from computer/device through serial connection. TX led Blinks when sending data to computer/device through serial connection. L led This led is connected to digital pin13 with a current limiter resistor (that doesn’t affect pin13). Useful to test sketches. It is normal to blink when bootloading too.

POWER PINOUT

6 pin header

RST pin

Makes Atmega reset when connected to GND. Useful for Shield Boards, or to connect external reset.

NC pin

This pin is not connected in Arduino S3v3. Arduino Diecimila has a 3.3 volts pin in the same position.

+9v pin

When Arduino DC1 is powered (with battery or DC adaptor), this pin is used as Vout, with the same voltage supplied on DC1 (see DC1), minus 0,7 volts. The total supplied current depends on external power source capacity When Arduino DC1 is not powered, +9v. pin can be used as Vin, connecting it to a external regulated power source (+7 to +20 volts) and connecting 0v. pin to external power source GND. In this case,

+5v pin

It can be used as Vout, supplying +5 volts. +5v. pin When Arduino DC1 is powered (with battery or DC adaptor), +5v. pin supplies +5 volts as a Vout pin. The total supplied current depends on Voltage Regulator (7805 supplies up to 1A). This applies only to +5v. pin: Atmega in/out pins only supplies max. 40mA on each pin. When Arduino DC1 is not powered, this pin can be used as Vin, connecting it to a regulated +5v. and connecting 0v. pin to power source GND. In this case, +9v. pin is inactive. 0v. pin (GND) Two 0v. pins between +5v. and +9v. / One

0v. pin

Beside AREF pin. When Arduino DC1 is powered, 0v. pin supplies 0 volts reference (GND) for +5v. pin and +9v. pin. When DC1 is not powered, and Arduino is powered through +5v. pin or +9v. pin, 0v. pin must be used as GND reference, connecting it to the external power source GND.

GND pin

see 0v. pin (GND).

AREF pin

The AREF can be set to AVcc (default), internal 2.56 volts (Atmega8), internal 1.1 volts (Atmega168), or external AREF. In case of AVcc or internal AREF, AREF pin can be used to

attach na external capacitor to decouple the signal, for better noise performance. In case of external AREF, AREF pin is used to attach the external reference voltage. Remember that it is necessary to change de fuses (wiring.c file), and re-upload sketch, before connecting external voltage to AREF

SOFTWARE TIPS

When bootloading na Atmega8 chip with Arduino 0010, there is a command (-i800) that makes bootloader delay 10 minutes. So, if you need to use bootloader, use command line instead of IDE, removing "–i800" command and adding "–F" command, or use Arduino 0007 IDE. To upload sketches Arduino 0010 works fine.

ARDUINO S3v3 NEW FEATURES

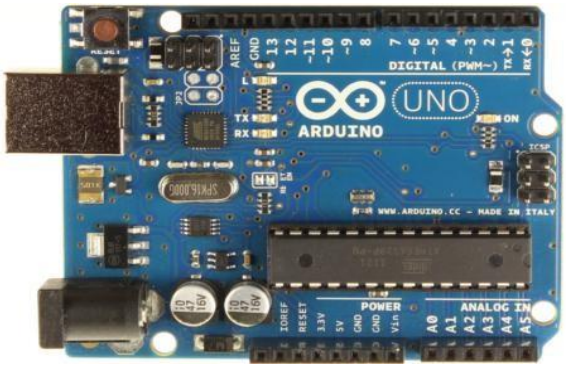
Full compatible with Shield Boards (Version 2 is the only Arduino Board not compatible with Shield Boards because of ICSP header wrong position, and tall components);

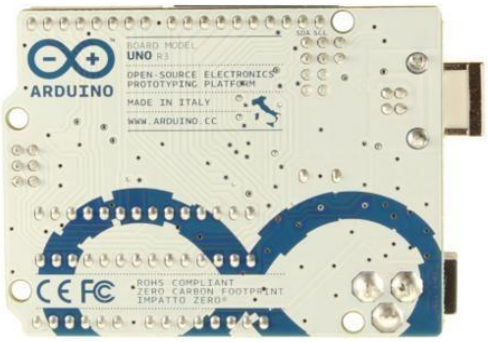
- A Vcc LP filter to reduce noise level on ADC;
- auto reset feature;
- auto reset enable/disable jumper, to avoid not desired reseting;
- arduino Diecimila compatible reset pin;
- pin13 onboard led, with current limiter resistor;
- TX and RX onboard leds;
- power led with appropriate current limiter resistor (less 20mA of comsumption);
- jumper to disable serial communication and to enable RX external pull down resistor, to avoid "RX floating error". This feature allows to use digital pin0 and pin1 as a normal pin, when serial communication is not needed;
- all similar components (diodes, transistors, leds, capacitors) has the same board orientation (to makes easier to mount with less mistakes);
- no wires between pads, more space between wires, larger wires, larger pads (better for etching, soldering and drilling, with no short circuits, soldering bridges or open wires in corrosion);
- only 3 wire bridges;
- electrolitic capacitor (in serial to TTL circuit) changed to bipolar type (to avoid inverted voltage problem when serial cable is not connected);
- All jumpers are right angle type, to allow Shield Boards use.

7. COMPONENTS USED FOR THE PROJECT :

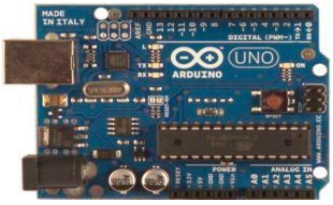
ARDUIINO UNO R3 BOARD :

Arduino Uno





Arduino Uno R3 Front *Arduino Uno R3 Back*



Arduino Uno R2 Front *Arduino Uno SMD* *Arduino Uno Front* *Arduino Uno Back*

Overview

The Arduino Uno is a microcontroller board based on the ATmega328 (datasheet). It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz ceramic resonator, a USB connection, a power jack, an ICSP header, and a reset button. It contains everything needed to support the microcontroller; simply connect it to a computer with a USB cable or power it with a AC-to-DC adapter or battery to get started.

The Uno differs from all preceding boards in that it does not use the FTDI USB-to-serial driver chip. Instead, it features the Atmega16U2 (Atmega8U2 up to version R2) programmed as a USB-to-serial converter.

Revision 2 of the Uno board has a resistor pulling the 8U2 HWB line to ground, making it easier to put into DFU mode.

Revision 3 of the board has the following new features:

1. 1.0 pinout: added SDA and SCL pins that are near to the AREF pin and two other new pins placed near to the RESET pin, the IOREF that allow the shields to adapt to the voltage provided from the board. In future, shields will be compatible both with the board that use the AVR, which operate with 5V and with the Arduino Due that operate with 3.3V. The second one is a not connected pin, that is reserved for future purposes.
2. Stronger RESET circuit.
3. Atmega 16U2 replace the 8U2.

"Uno" means one in Italian and is named to mark the upcoming release of Arduino 1.0. The Uno and version 1.0 will be the reference versions of Arduino, moving forward. The Uno is the latest in a series of USB Arduino boards, and the reference model for the Arduino platform; for a comparison with previous versions, see the index of Arduino boards.

Summary :

Microcontroller	ATmega328
Operating Voltage	5V
Input Voltage (recommended)	7-12V
Input Voltage (limits)	6-20V
Digital I/O Pins	14 (of which 6 provide PWM output)
Analog Input Pins	6
DC Current per I/O Pin	40 mA
DC Current for 3.3V Pin	50 mA
Flash Memory	32 KB (ATmega328) of which 0.5 KB used by bootloader
SRAM	2 KB (ATmega328)
EEPROM	1 KB (ATmega328)
Clock Speed	16 MHz

Schematic & Reference Design

EAGLE files: arduino-uno-Rev3-reference-design.zip (NOTE: works with Eagle 6.0 and newer)
Schematic: arduino-uno-Rev3-schematic.pdf

Note: The Arduino reference design can use an Atmega8, 168, or 328, Current models use an ATmega328, but an Atmega8 is shown in the schematic for reference. The pin configuration is identical on all three processors.

Power

The Arduino Uno can be powered via the USB connection or with an external power supply. The power source is selected automatically.

External (non-USB) power can come either from an AC-to-DC adapter (wall-wart) or battery. The adapter can be connected by plugging a 2.1mm center-positive plug into the board's power jack. Leads from a battery can be inserted in the Gnd and Vin pin headers of the POWER connector.

The board can operate on an external supply of 6 to 20 volts. If supplied with less than 7V, however, the 5V pin may supply less than five volts and the board may be unstable. If using more than 12V, the voltage regulator may overheat and damage the board. The recommended range is 7 to 12 volts.

The power pins are as follows:

1. **VIN.** The input voltage to the Arduino board when it's using an external power source (as opposed to 5 volts from the USB connection or other regulated power source). You can supply voltage through this pin, or, if supplying voltage via the power jack, access it through this pin.
2. **5V.**This pin outputs a regulated 5V from the regulator on the board. The board can be supplied with power either from the DC power jack (7 - 12V), the USB connector (5V), or the VIN pin of the board (7-12V). Supplying voltage via the 5V or 3.3V pins bypasses the regulator, and can damage your board. We don't advise it.
3. **3V3.** A 3.3 volt supply generated by the on-board regulator. Maximum current draw is 50 mA.
4. **GND.** Ground pins.

Memory

The ATmega328 has 32 KB (with 0.5 KB used for the bootloader). It also has 2 KB of SRAM and 1 KB of EEPROM (which can be read and written with the EEPROM library).

Input and Output

Each of the 14 digital pins on the Uno can be used as an input or output, using pinMode(), digitalWrite(), and digitalRead() functions. They operate at 5 volts. Each pin can provide or receive a maximum of 40 mA and has an internal pull-up resistor (disconnected by default) of 20-50 kOhms. In addition, some pins have specialized functions:

1. **Serial: 0 (RX) and 1 (TX).** Used to receive (RX) and transmit (TX) TTL serial data. These pins are connected to the corresponding pins of the ATmega8U2 USB-to-TTL Serial chip.

2. **External Interrupts: 2 and 3.** These pins can be configured to trigger an interrupt on a low value, a rising or falling edge, or a change in value. See the attachInterrupt() function for details.

3. **PWM: 3, 5, 6, 9, 10, and 11.** Provide 8-bit PWM output with the analogWrite() function. **SPI: 10 (SS), 11 (MOSI), 12 (MISO), 13 (SCK).** These pins support SPI communication using the SPI library.

4. **LED: 13.** There is a built-in LED connected to digital pin 13. When the pin is HIGH value, the LED is on, when the pin is LOW, it's off.

5. The Uno has 6 analog inputs, labeled A0 through A5, each of which provide 10 bits of resolution (i.e. 1024 different values). By default they measure from ground to 5 volts, though is it possible to change the upper end of their range using the AREF pin and the analogReference() function. Additionally, some pins have specialized functionality:

6. **TWI: A4 or SDA pin and A5 or SCL pin.** Support TWI communication using the Wire library.

7. There are a couple of other pins on the board:

8. **AREF.** Reference voltage for the analog inputs. Used with analogReference().

9. **Reset.** Bring this line LOW to reset the microcontroller. Typically used to add a reset button to shields which block the one on the board.

See also the mapping between Arduino pins and ATmega328 ports. The mapping for the Atmega8, 168, and 328 is identical.

Communication

The Arduino Uno has a number of facilities for communicating with a computer, another Arduino, or other microcontrollers. The ATmega328 provides UART TTL (5V) serial communication, which is available on digital pins 0 (RX) and 1 (TX). An ATmega16U2 on the board channels this serial communication over USB and appears as a virtual com port to software on the computer. The '16U2 firmware uses the standard USB COM drivers, and no external driver is needed.

However, on Windows, a .inf file is required. The Arduino software includes a serial monitor which allows simple textual data to be sent to and from the Arduino board. The RX and TX LEDs on the board will flash when data is being transmitted via the USB-to-serial chip and USB connection to the computer (but not for serial communication on pins 0 and 1).

A SoftwareSerial library allows for serial communication on any of the Uno's digital pins.

The ATmega328 also supports I2C (TWI) and SPI communication. The Arduino software includes a Wire library to simplify use of the I2C bus; see the documentation for details. For SPI communication, use the SPI library.

Programming

The Arduino Uno can be programmed with the Arduino software (download). Select "Arduino Uno from the **Tools > Board** menu (according to the microcontroller on your board). For details, see the reference and tutorials.

The ATmega328 on the Arduino Uno comes preburned with a bootloader that allows you to upload new code to it without the use of an external hardware programmer. It communicates using the original STK500 protocol (reference, C header files).

You can also bypass the bootloader and program the microcontroller through the ICSP (In-Circuit Serial Programming) header; see these instructions for details.

The ATmega16U2 (or 8U2 in the rev1 and rev2 boards) firmware source code is available . The ATmega16U2/8U2 is loaded with a DFU bootloader, which can be activated by:

1. On Rev1 boards: connecting the solder jumper on the back of the board (near the map of Italy) and then resetting the 8U2.
2. On Rev2 or later boards: there is a resistor that pulling the 8U2/16U2 HWB line to ground, making it easier to put into DFU mode.

You can then use Atmel's FLIP software (Windows) or the DFU programmer (Mac OS X and Linux) to load a new firmware. Or you can use the ISP header with an external programmer (overwriting the DFU bootloader). See this user-contributed tutorial for more information.

Automatic (Software) Reset

Rather than requiring a physical press of the reset button before an upload, the Arduino Uno is designed in a way that allows it to be reset by software running on a connected computer. One of the hardware flow control lines (DTR) of the ATmega8U2/16U2 is connected to the reset line of the ATmega328 via a 100 nanofarad capacitor. When this line is asserted (taken low), the reset line drops long enough to reset the chip. The Arduino software uses this capability to allow you to upload code by simply pressing the upload button in the Arduino environment. This means that the bootloader can have a shorter timeout, as the lowering of DTR can be well-coordinated with the start of the upload.

This setup has other implications. When the Uno is connected to either a computer running Mac OS X or Linux, it resets each time a connection is made to it from software (via USB). For the following half- second or so, the bootloader is running on the Uno.

While it is programmed to ignore malformed data (i.e. anything besides an upload of new code),

it will intercept the first few bytes of data sent to the board after a connection is opened. If a sketch running on the board receives one-time configuration or other data when it first starts,

make sure that the software with which it communicates waits a second after opening the connection and before sending this data.

The Uno contains a trace that can be cut to disable the auto-reset. The pads on either side of the trace can be soldered together to re-enable it. It's labeled "RESET-EN". You may also be able to disable the auto-reset by connecting a 110 ohm resistor from 5V to the reset line; see this forum thread for details.

USB Overcurrent Protection

The Arduino Uno has a resettable polyfuse that protects your computer's USB ports from shorts and overcurrent. Although most computers provide their own internal protection, the fuse provides an extra layer of protection. If more than 500 mA is applied to the USB port, the fuse will automatically break the connection until the short or overload is removed.

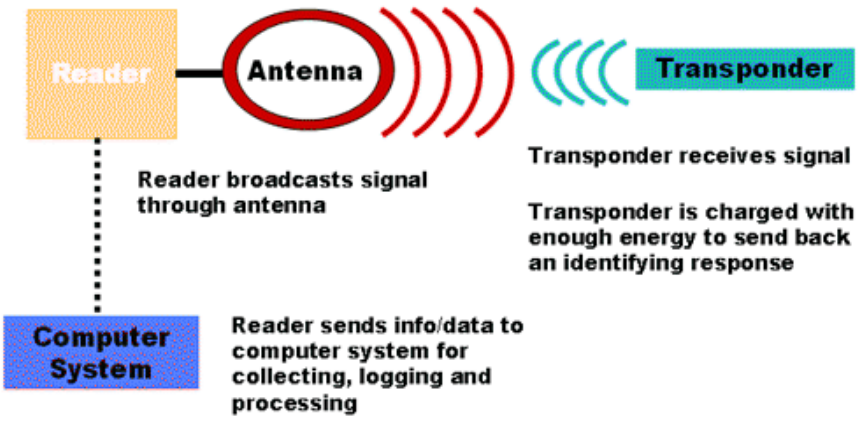
Physical Characteristics

The maximum length and width of the Uno PCB are 2.7 and 2.1 inches respectively, with the USB connector and power jack extending beyond the former dimension. Four screw holes allow the board to be attached to a surface or case. Note that the distance between digital pins 7 and 8 is 160 mil (0.16"), not an even multiple of the 100 mil spacing of the other pins.

RFID TAGS :

RFID (radio frequency identification reader)

A radio frequency identification reader (RFID reader) is a device used to gather information from an RFID tag, which is used to track individual objects. ... RFID is a technology similar in theory to bar codes. However, the RFID tag does not have to be scanned directly, nor does it require line-of-sight to a reader



RFID is a technology similar in theory to bar codes. However, the RFID tag does not have to be scanned directly, nor does it require line-of-sight to a reader. The RFID tag it must be within the range of an RFID reader, which ranges from 3 to 300 feet, in order to be read. RFID technology allows several items to be quickly scanned and enables fast identification of a particular product, even when it is surrounded by several other items.

RFID tags have not replaced bar codes because of their cost and the need to individually identify every item.

RFID or Radio Frequency Identification System is a technology based identification system which helps identifying objects just through the tags attached to them, without requiring any light of sight between the tags and the tag reader. All that is needed is radio communication between the tag and the reader.

A RFID tag:

It consists of a silicon microchip attached to a small antenna and mounted on a substrate and encapsulated in different materials like plastic or glass veil and with an adhesive on the back side to be attached to objects.

1. **A reader:** It consists of a scanner with antennas to transmit and receive signals and is responsible for communication with the tag and receives the information from the tag.
2. **A Processor or a Controller**: It can be a host computer with a Microprocessor or a microcontroller which receives the reader input and process the data.

Types of RFID Systems:

Active RFID system:

These are systems where the tag has its own power source like any external power supply unit or a battery. The only constraint being the life time of the power devices. These systems can be used for larger distances and to track high value goods like vehicles.

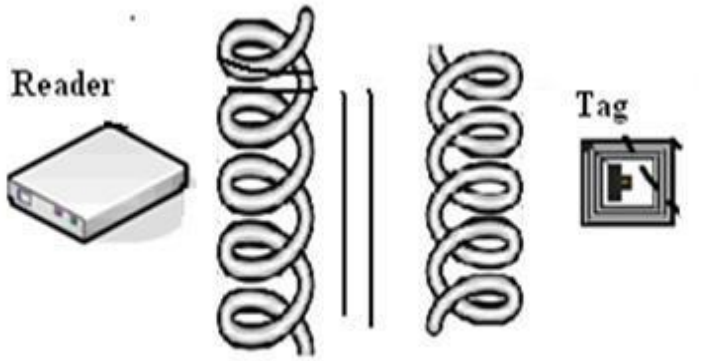
Passive RFID system:

These are systems where the tag gets power through the transfer of power from a reader antenna to the tag antenna. They are used for short range transmission

How the Passive RFID System Works:

The tag can be powered either using inducting coupling method or through EM wave capture method. Lets us have a brief knowledge about the system using these two methods

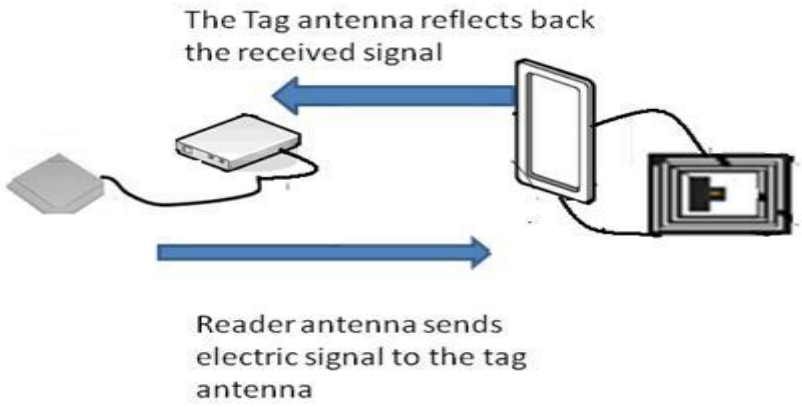
Virtue of Faraday's law of induction. The EMF causes a flow of current in the coil, thus producing a magnetic field around it. By the virtue of Lenz law, the magnetic field of the tag coil opposes the reader's magnetic field and there will be a subsequent increase in the current through the reader coil. The reader intercepts this as the load information. This system is suitable for very short distance communication. The AC voltage appearing across the tag coil is converted to DC using rectifier and filter arrangement.



Reader Coil Inductively coupled to tag coil

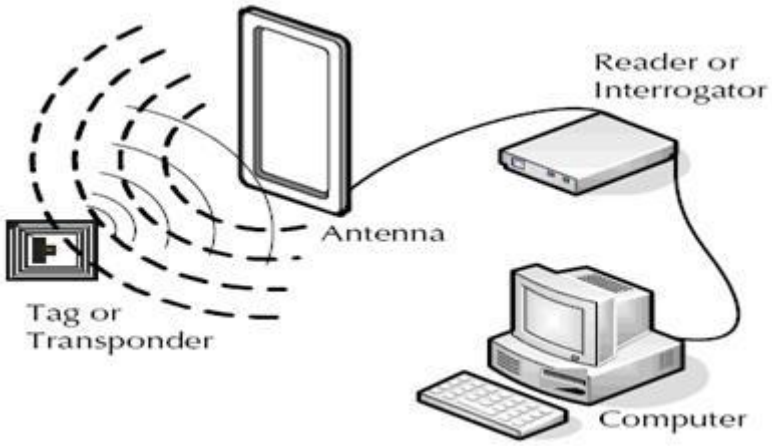
A Passive RFID system using EM wave propagation method:

The antenna present in the reader transmits electromagnetic waves which are received by the antenna present in the tag as potential difference across the dipole. This voltage is rectified and filtered to get the DC power. The receiver antenna is kept at different impedance which causes it to reflect a part of the received signal. This reflected signal is received by the reader and monitored accordingly.



Active RFID System Works:

In the active RFID system, the reader sends signal to the tag using an antenna. The tag receives this information and resends this information along with the information in its memory. The reader receives this signal and transmits to the processor for further processing.



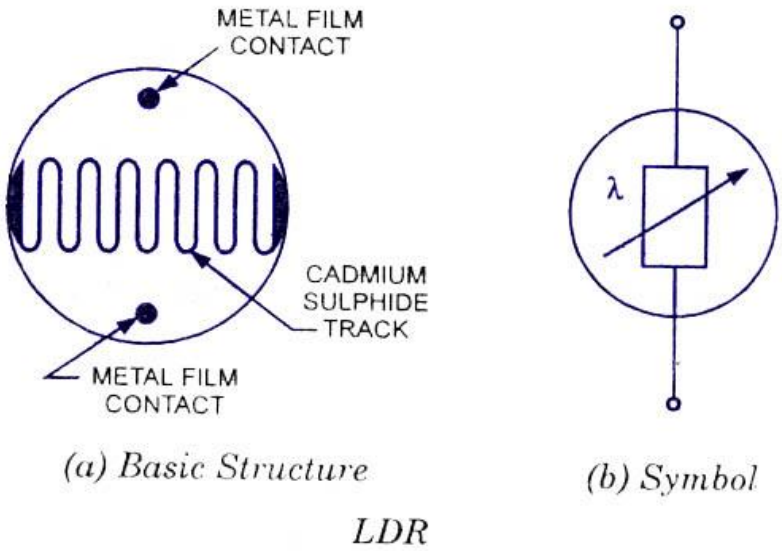
Application

1. Passports
2. Smart cards
3. Airplane luggage
4. Toll booth passes
5. Home appliances
6. Merchandise tags
7. Animal and pet tags
8. Automobile key-and-lock
9. Monitoring heart patients
10. Pallet tracking for inventory
11. Telephone and computer networks
12. Operation of spacecraft and satellites

LDR SENSOR :

Light Dependent Resistor (LDR)

A *Light Dependent Resistor* (LDR) or a photo resistor is a device whose resistivity is a function of the incident electromagnetic radiation. Hence, they are light sensitive devices. They are also called as photo conductors, photo conductive cells or simply photocells. They are made up of semiconductor materials having high resistance. There are many different symbols used to indicate a **LDR**, one of the most commonly used symbol is shown in the figure below. The arrow indicates light falling on it.

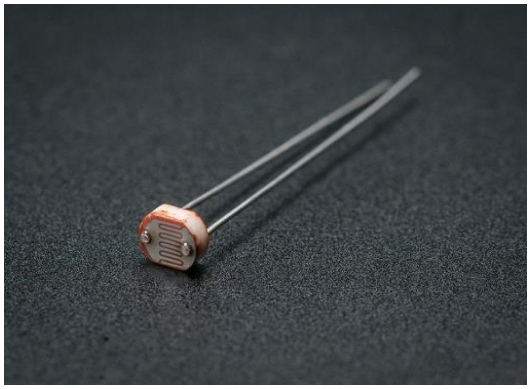


Construction of a Photocell

The structure of a light dependent resistor consists of a light sensitive material which is deposited on an insulating substrate such as ceramic. The material is deposited in zigzag pattern in order to obtain the desired resistance and power rating. This zigzag area separates the metal deposited areas into two regions. Then the ohmic contacts are made on the either sides of the area. The resistances of these contacts should be as less as possible to make sure that the resistance mainly changes due to the effect of light only. Materials normally used are cadmium sulphide, cadmium selenide, indium antimonide and cadmium sulphonide. The use of lead and cadmium is avoided as they are harmful to the environment.

Characteristics of LDR

LDR's are light dependent devices whose resistance is decreased when light falls on them and that is increased in the dark. When a light dependent resistor is kept in dark, its resistance is very high. This resistance is called as dark resistance. It can be as high as 1012 Ω and if the device is allowed to absorb light its resistance will be decreased drastically. If a constant voltage is applied to it and intensity of light is increased the current starts increasing. Figure below shows resistance vs. illumination curve for a particular LDR.



LDR characteristics

Photocells or LDR's are non-linear devices. There sensitivity varies with the wavelength of light incident on them. Some photocells might not at all response to a certain range of wavelengths. Based on the material used different cells have different spectral response curves.

When light is incident on a photocell it usually takes about 8 to 12 ms for the change in resistance to take place, while it takes one or more seconds for the resistance to rise back again to its initial value after removal of light. This phenomenon is called as resistance recovery rate. This property is used in audio compressors.

Also, LDR's are less sensitive than photo diodes and phototransistor. (A photo diode and a photocell (LDR) are not the same, a photo-diode is a pn junction semiconductor device that converts light to electricity, whereas a photocell is a passive device, there is no pn junction in this nor it "converts" light to electricity).

Types of Light Dependent Resistors

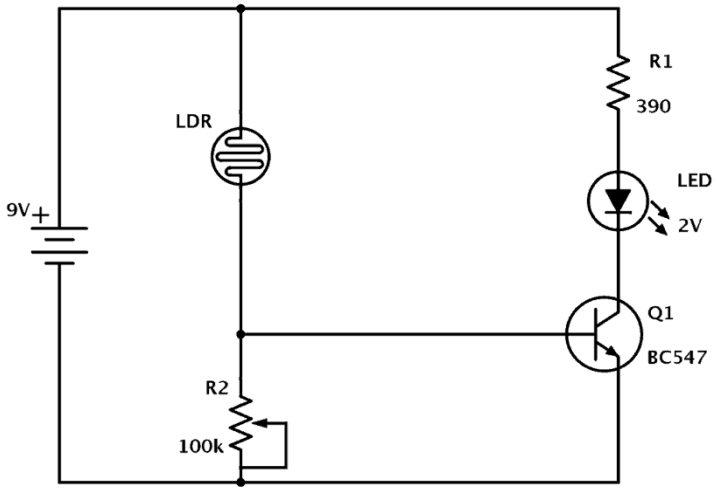
Based on the materials used they are classified as:

1. Intrinsic photo resistors (Un doped semiconductor): These are made of pure semiconductor materials such as silicon or germanium. Electrons get excited from valance band to conduction band when photons of enough energy fall on it and number charge carriers is increased.
2. Extrinsic photo resistors: These are semiconductor materials doped with impurities which are called as dopants. Theses dopants create new energy bands above the valence band which are filled with electrons. Hence this reduces the band gap and less energy is required in exciting them. Extrinsic photo resistors are generally used for long wavelengths.

Working Principle of LDR

A light dependent resistor works on the principle of photo conductivity. Photo conductivity is an optical phenomenon in which the materials conductivity is increased when light is absorbed by the material. When light falls i.e. when the photons fall on the device, the electrons in the valence band of the semiconductor material are excited to the conduction band.

These photons in the incident light should have energy greater than the band gap of the semiconductor material to make the electrons jump from the valence band to the conduction band. Hence when light having enough energy strikes on the device, more and more electrons are excited to the conduction band which results in large number of charge carriers. The result of this process is more and more current starts flowing through the device when the circuit is closed and hence it is said that the resistance of the device has been decreased.



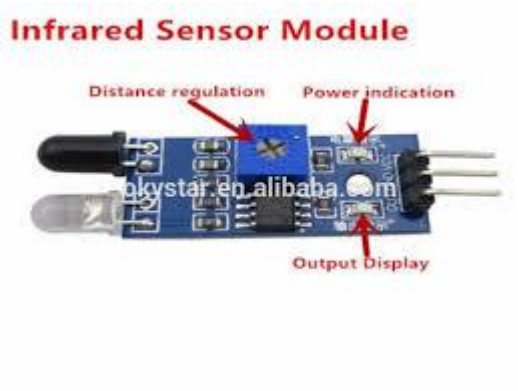
Applications

1. Camera light meter
2. Street lamps
3. Alarm clock
4. Burglar alarm circuits
5. Light intensity meters
6. Clock radios
7. Light beam alarms
8. Reflective smoke alarms
9. Outdoor clocks.

IR SENSOR:

IR (Infrared Sensor)

IR sensor is a simple electronic device which emits and detects IR radiation in order to find out certain objects/obstacles in its range. Some of its features are heat and motion sensing.



IR sensors use infrared radiation of wavelength between 0.75 to 1000µm which falls between visible and microwave regions of electromagnetic spectrum. IR region is not visible to human eyes. Infrared spectrum is categorized into three regions based on its wavelength i.e. Near Infrared, Mid Infrared, Far Infrared.

Wavelength Regions of Infrared Spectrum

1. Near IR – 0.75µm to 3 µm
2. Mid IR – 3 µm to 6 µm
3. Far IR – > 6 µm

Working Principle of Infrared Sensor

Infrared Sensors works on three fundamental Physics laws:

IR Transmitter

IR Transmitter acts as source for IR radiation. According to Plank's Radiation Law, every object is a source of IR radiation at temp T above 0 Kelvin. In most cases black body radiators, tungsten lamps, silicon carbide, infrared lasers, LEDs of infrared wavelength are used as sources.

Transmission Medium

As the name suggests, Transmission Medium provides passage for the radiation to reach from IR Transmitter to IR Receiver. Vacuum, atmosphere and optical fibers are used as medium.

IR receiver

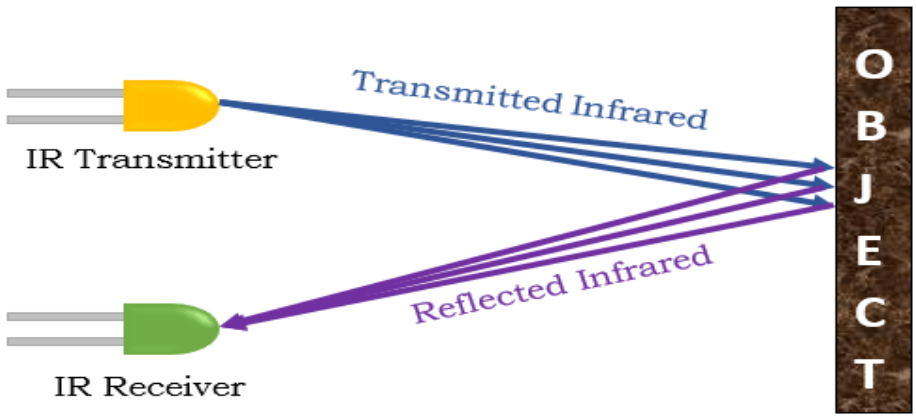
Generally IR receivers are photo diode and photo transistors. They are capable of detecting infrared radiation. Hence IR receiver is also called as IR detector. Variety of receivers are available based on wavelength, voltage and package.

IR Transmitter and Receivers are selected with matching parameters. Some of deciding specifications of receivers are photosensitivity or responsivity, noise equivalent power and detectivity.

How Infrared Sensor Works

An Infrared Sensor works in the following sequence:

1. IR source (transmitter) is used to emit radiation of required wavelength.
2. This radiation reaches the object and is reflected back.
3. The reflected radiation is detected by the IR receiver.
4. The IR Receiver detected radiation is then further processed based on its intensity. Generally, IR Receiver output is small and amplifiers are used to amplify the detected signal.



Incidence in an IR Detection System may be direct or indirect. In case of Direct Incidence, there is no hindrance in between transmitter and receiver. Whereas, in Indirect Incidence IR Transmitter and Receiver are kept side by side and the object is in front of them.

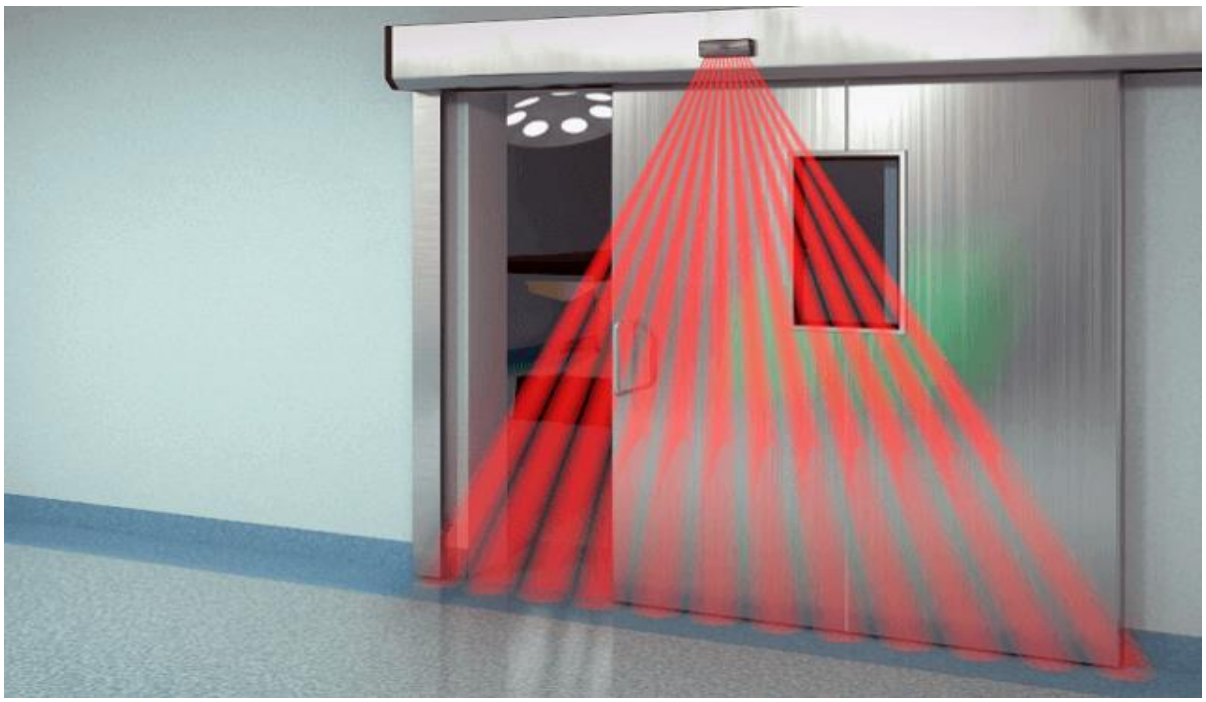
Types of Infrared Sensor

IR sensors can be classified in two types based on presence of IR source:

- Active Infrared Sensor
- Passive Infrared Sensor

Active Infrared Sensor

Active Infrared Sensor contains both transmitter and receiver. Most of the cases LED or laser diode is used as source. LED for non-imaging IR sensor and laser diode for imaging IR sensor are used.



Active IR Sensor works by radiating energy, received and detected by detector and further processed by signal processor in order to fetch information required.

Examples of Active IR Sensor: Break Beam Sensor, Reflectance Sensor.

Passive Infrared Sensor

Passive Infrared Sensor contains detectors alone. There won't be a transmitter component.

These type of sensors use object as IR source/ transmitter. Object radiates energy and it is detected by IR receivers. A Signal processor is then used to interpret the signal to fetch information required.

Example of Passive IR Sensor: Thermocouple-Thermopile, Bolometer, Pyro-Electric Detector, etc.

There are two types of Passive Infrared Sensor:

1. Thermal Infrared Sensor
2. Quantum Infrared Sensor

Thermal Infrared Sensor :

Thermal Infrared sensors are independent of wavelength. They use heat as energy source.

Thermal detectors are slow with their detection time and response time.

Quantum Infrared Sensor

Quantum Infrared Sensor are dependent on wavelengths. They have high detection time and response time. These type of IR sensors require frequent cooling for precise measurement.

Applications of Infrared Sensor

IR sensors have found their applications in most of today's equipment. Following are the list of sensors which are named after its usage.

Proximity Sensor

These are used in smart phones to find distance of object. They use principle called Reflective Indirect Incidence. Radiation transmitted by transmitter is received by receiver after being reflected from object. Distance is calculated based on the intensity of radiation received.

Item Counter

This use direct incidence method to count the items. Constant radiation is maintained in between transmitter and receiver. As soon as object cuts the radiation, item is detected and count is increased. The same count is shown on display system.

Burglar Alarm

This is one of widely and commonly used sensor application. It is another example for direct incidence method.

It works similar to item counter, where transmitter and receiver are kept on both the sides of door frame. Constant radiation is maintained between transmitter and receiver, whenever object crosses path alarm starts off.

Radiation Thermometers

It is one of key application of Infrared sensors. Working of radiation thermometer depends on temperature and type of object.

These have faster response and easy pattern measurements. They can do measurement without direct contact of object.

Human Body Detection

This method is used in intrusion detection, auto light switches, etc. Intrusion alarm system sense temperature of human body.

If the temperature is more than threshold value, it sets on the alarms. It uses electromagnetic system which is suitable for human body in order to protect it from unwanted harmful radiations.

Gas Analyzers

Gas Analyzers are used to measure gas density by using absorption properties of gas in IR region. Dispersive and Non Dispersive types of gas analyzers are available.

Other Applications

IR sensors are also used in IR imaging devices, optical power meters, sorting devices, missile guidance, remote sensing, flame monitors, moisture analyzers, night vision devices, infrared astronomy, rail safety, etc.

IR Sensor Module Features

1. 5VDC Operating voltage
2. I/O pins are 5V and 3.3V compliant
3. Range: Up to 20cm
4. Adjustable Sensing range
5. Built-in Ambient Light Sensor
6. 20mA supply current
7. Mounting hole

Advantages of Infrared Sensor

The advantages of Infrared Sensor are:

1. Their low power requirements make them suitable for most electronic devices such as laptops, telephones, PDAs.
2. They are capable of detecting motion in presence/ absence of light almost with same reliability.
3. They do not require contact with object to for detection.
4. There is no leakage of data due to beam directionality IR radiation.
5. They are not affected by corrosion or oxidation.
6. They have very strong noise immunity.

Disadvantages of Infrared Sensor

The disadvantages of Infrared Sensor are:

1. Required Line of sight.
2. Get blocked by common objects.
3. Limited range.
4. Can be affected by Environmental conditions such as rain, fog, dust, pollution.
5. Transmission Data rate is slow.
6.

Applications

1. Obstacle Detection
2. Industrial safety devices
3. Wheel encode

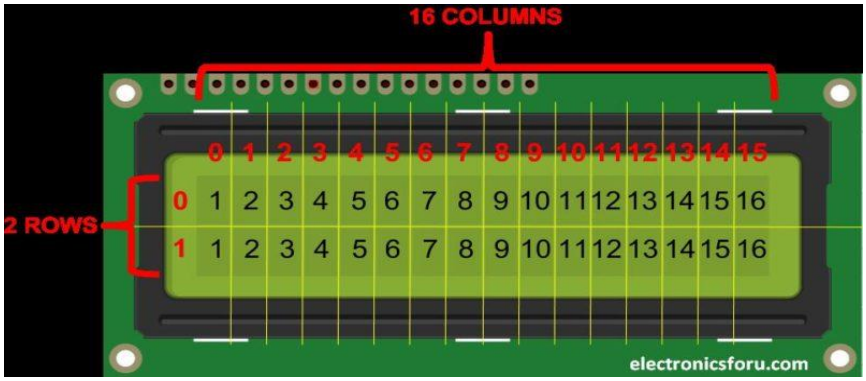
LCD (Liquid Crystal Display) :

LCD (Liquid Crystal Display) screen is an electronic display module and find a wide range of applications. A 16×2 LCD display is very basic module and is very commonly used in various devices and circuits. These modules are preferred over seven segments and other multi segment LEDs. The reasons being: LCDs are economical; easily programmable; have no limitation of displaying special & even custom characters (unlike in seven segments), animations and so on.

A 16×2 LCD means it can display 16 characters per line and there are 2 such lines. In this LCD each character is displayed in 5×7 pixel matrix. This LCD has two registers, namely, Command and Data.

The command register stores the command instructions given to the LCD. A command is an instruction given to LCD to do a predefined task like initializing it, clearing its screen, setting the cursor position, controlling display etc. The data register stores the data to be displayed on the

LCD. The data is the ASCII value of the character to be displayed on the LCD. Click to learn more about internal structure of a LCD.



How LCDs are Constructed?

1. The basic structure of LCD should be controlled by changing the applied current.
2. We must use a polarized light.
3. Liquid crystal should able be to control both of the operation to transmit or can also able to change the polarized light.

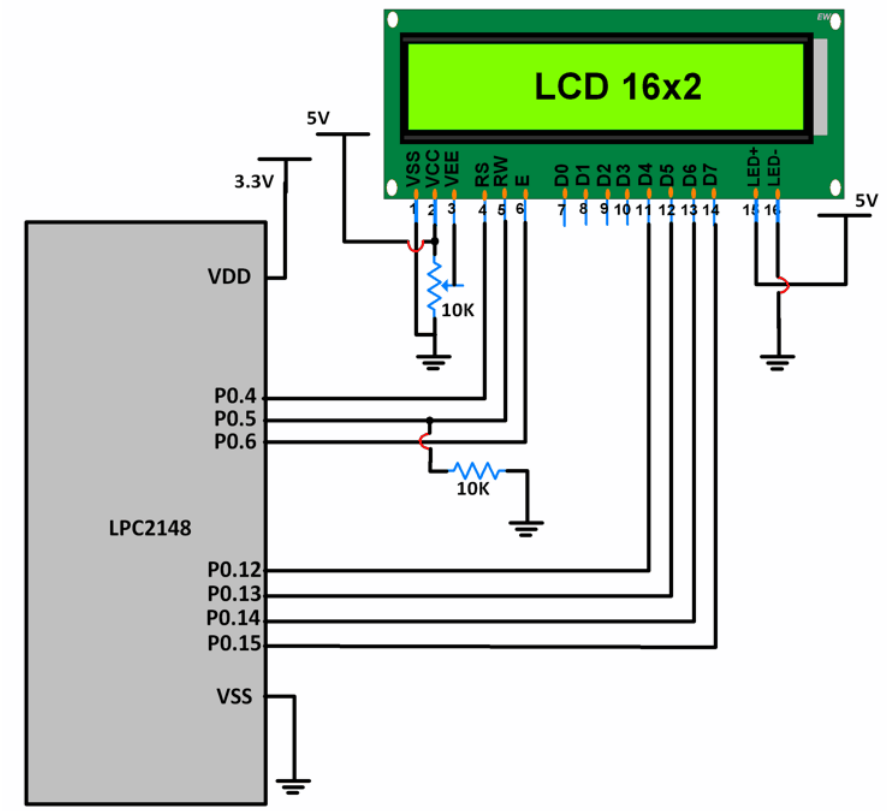
As mentioned above that we need to take two polarized glass pieces filter in the making of the liquid crystal. The glass which does not have a polarized film on the surface of it must be rubbed with a special polymer which will create microscopic grooves on the surface of the polarized glass filter. The grooves must be in the same direction of the polarized film. Now we have to add a coating of pneumatic liquid phase crystal on one of the polarized filter of the polarized glass. The microscopic channel cause the first layer molecule to align with filter orientation. When the right angle appears at the first layer piece, we should add a second piece of glass with the polarized film. The first filter will be naturally polarized as the light strikes it at the starting stage.

Thus the light travels through each layer and guided on the next with the help of molecule. The molecule tends to change its plane of vibration of the light in order to match their angle. When the light reaches to the far end of the liquid crystal substance, it vibrates at the same angle

as that of the final layer of the molecule vibrates. The light is allowed to enter into the device only if the second layer of the polarized glass matches with the final layer of the molecule.

Working

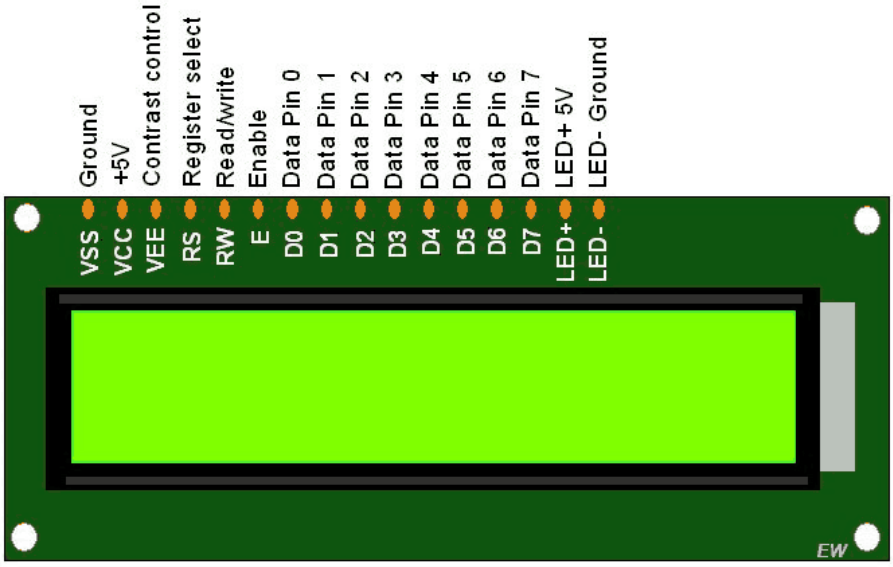
The principle behind the LCD's is that when an electrical current is applied to the liquid crystal molecule, the molecule tends to untwist. This causes the angle of light which is passing through the molecule of the polarized glass and also cause a change in the angle of the top polarizing filter. As a result a little light is allowed to pass the polarized glass through a particular area of the LCD. Thus that particular area will become dark compared to other. The LCD works on the principle of blocking light. While constructing the LCD's, a reflected mirror is arranged at the back. An electrode plane is made of indium-tin oxide which is kept on top and a polarized glass with a polarizing film is also added on the bottom of the device. The complete region of the LCD has to be enclosed by a common electrode and above it should be the liquid crystal matter.



Next comes to the second piece of glass with an electrode in the form of the rectangle on the bottom and, on top, another polarizing film. It must be considered that both the pieces are kept at right angles. When there is no current, the light passes through the front of the LCD it will be reflected by the mirror and bounced back. As the electrode is connected to a battery the current from it will cause the liquid crystals between the common-plane electrode and the electrode

shaped like a rectangle to untwist. Thus the light is blocked from passing through. That particular rectangular area appears blank.

16X2 LCD pinout diagram



Sr. No	Pin No.	Pin Name	Pin Type	Pin Description	Pin Connection
1	Pin 1	Ground	Source Pin	This is a ground pin of LCD	Connected to the ground of the MCU/ Power source
2	Pin 2	VCC	Source Pin	This is the supply voltage pin of LCD	Connected to the supply pin of Power source
3	Pin 3	V0/VEE	Control Pin	Adjusts the contrast of the LCD.	Connected to a variable POT that can source 0-5V
4	Pin 4	Register Select	Control Pin	Toggles between Command/Data Register	Connected to a MCU pin and gets either 0 or 1. 0 -> Command Mode 1-> Data Mode

5	Pin 5	Read/Write	Control Pin	Toggles the LCD between Read/Write Operation	Connected to a MCU pin and gets either 0 or 1. 0 -> Write Operation 1-> Read Operation
6	Pin 6	Enable	Control Pin	Must be held high to perform Read/Write Operation	Connected to MCU and always held high.
7	Pin 7-14	Data Bits (0-7)	Data/Command Pin	Pins used to send Command or data to the LCD.	In 4-Wire Mode Only 4 pins (0-3) is connected to MCU In 8-Wire Mode All 8 pins(0-7) are connected to MCU
8	Pin 15	LED Positive	LED Pin	Normal LED like operation to illuminate the LCD	Connected to +5V
9	Pin 16	LED Negative	LED Pin	Normal LED like operation to illuminate the LCD connected with GND.	Connected to ground

Features of 16×2 LCD module

1. Operating Voltage is 4.7V to 5.3V
2. Current consumption is 1mA without backlight
3. Alphanumeric LCD display module, meaning can display alphabets and numbers
4. Consists of two rows and each row can print 16 characters.
5. Each character is build by a 5×8 pixel box
6. Can work on both 8-bit and 4-bit mode
7. It can also display any custom generated characters
8. Available in Green and Blue Backlight

Advantages

1. LCD's consumes less amount of power compared to CRT and LED
2. LCD's are consist of some microwatts for display in comparison to some mill watts for LED's
3. LCDs are of low cost
4. Provides excellent contrast
5. LCD's are thinner and lighter when compared to cathode ray tube and LED

Disadvantages

1. Require additional light sources
2. Range of temperature is limited for operation
3. Low reliability
4. Speed is very low
5. LCD's need an AC drive

Applications

1. Liquid crystal technology has major applications in the field of science and engineering as well on electronic devices.
2. Liquid crystal thermometer
3. Optical imaging
4. The liquid crystal display technique is also applicable in visualization of the radio frequency waves in the waveguide
5. Used in the medical applications

Buzzer

A buzzer or beeper is an audio signalling device, which may be mechanical, electromechanical, or piezoelectric (piezo for short). Typical uses of buzzers and beepers include alarm devices, timers, and confirmation of user input such as a mouse click or keystroke

How to use a Buzzer

A buzzer is a small yet efficient component to add sound features to our project/system. It is very small and compact 2-pin structure hence can be easily used on breadboard, Perf Board and even on PCBs which makes this a widely used component in most electronic applications.

There are two types are buzzers that are commonly available. The one shown here is a simple buzzer which when powered will make a Continuous Beeeeeeppp.... sound, the other type is called a readymade buzzer which will look bulkier than this and will produce a Beep. Beep. Beep. Sound due to the internal oscillating circuit present inside it. But, the one shown here is most widely used because it can be customised with help of other circuits to fit easily in our application.

This buzzer can be used by simply powering it using a DC power supply ranging from 4V to 9V. A simple 9V battery can also be used, but it is recommended to use a regulated +5V or +6V DC supply. The buzzer is normally associated with a switching circuit to turn ON or turn OFF the buzzer at required time and require interval.

Types

Electromechanical

Early devices were based on an electromechanical system identical to an electric bell without the metal gong. Similarly, a relay may be connected to interrupt its own actuating current, causing the contacts to buzz. Often these units were anchored to a wall or ceiling to use it as a sounding board. The word "buzzer" comes from the rasping noise that electromechanical buzzers made.

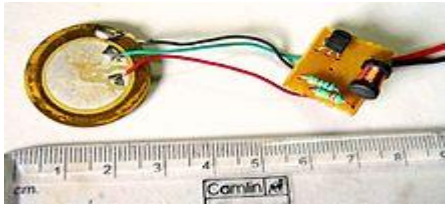
Mechanical

A joy buzzer is an example of a purely mechanical buzzer and they require drivers. Other examples of them are doorbells.

Piezoelectric

Piezoelectric disk beeper

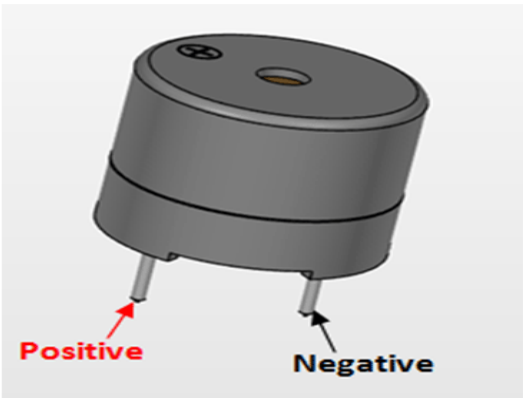
A piezoelectric element may be driven by an oscillating electronic circuit or other audio signal source, driven with a piezoelectric audio amplifier. Sounds commonly used to indicate that a button has been pressed are a click, a ring or a beep.



Interior of a readymade loudspeaker, showing a piezoelectric-disk-beeper (With 3 electrodes ... including 1 feedback-electrode (the central, small electrode joined with red wire in this photo), and an oscillator to self-drive the buzzer.

A piezoelectric buzzer/beeper also depends on acoustic cavity resonance or Helmholtz resonance to produce an audible beep

Buzzer Pin **Configuration**



Pin Number	Pin Name	Description
1	Positive	Identified by (+) symbol or longer terminal lead. Can be powered by 6V DC
2	Negative	Identified by short terminal lead. Typically connected to the ground of the circuit

Specifications

1. Rated Voltage: 6V DC
2. Operating Voltage: 4-8V DC
3. Rated current: <30mA
4. Sound Type: Continuous Beep
5. Resonant Frequency: ~2300 Hz
6. Small and neat sealed package
7. Breadboard and Perf board friendly

Application

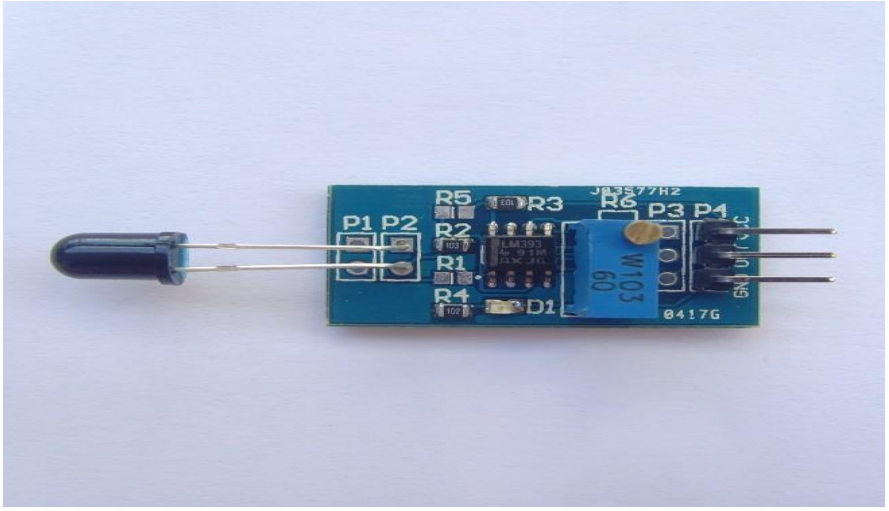
1. Alarming Circuits, where the user has to be alarmed about something
2. Communication equipments
3. Automobile electronics
4. Portable equipments, due to its compact size

FLAME SENSOR :

Fire/Flame Sensor Module

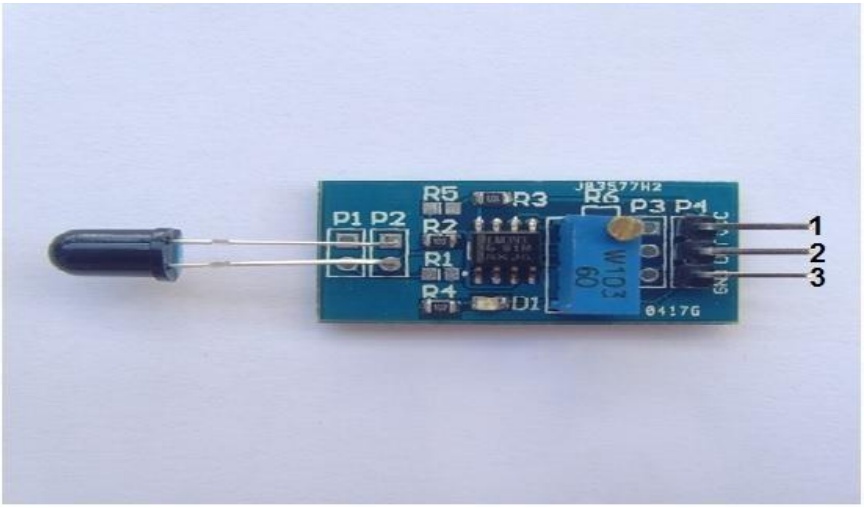
General Description:

Flame sensor is the most sensitive to ordinary light that is why its reaction is generally used as flame alarm purposes. This module can detect flame or wavelength in 760 nm to 1100 nm range of light source. Small plate output interface can and single- chip can be directly connected to the microcomputer IO port. The sensor and flame should keep a certain distance to avoid high temperature damage to the sensor. The shortest test distance is 80 cm, if the flame is bigger, test it with farther distance. The detection angle is 60 degrees so the flame spectrum is especially sensitive. The detection angle is 60 degrees so the flame spectrum is especially sensitive.



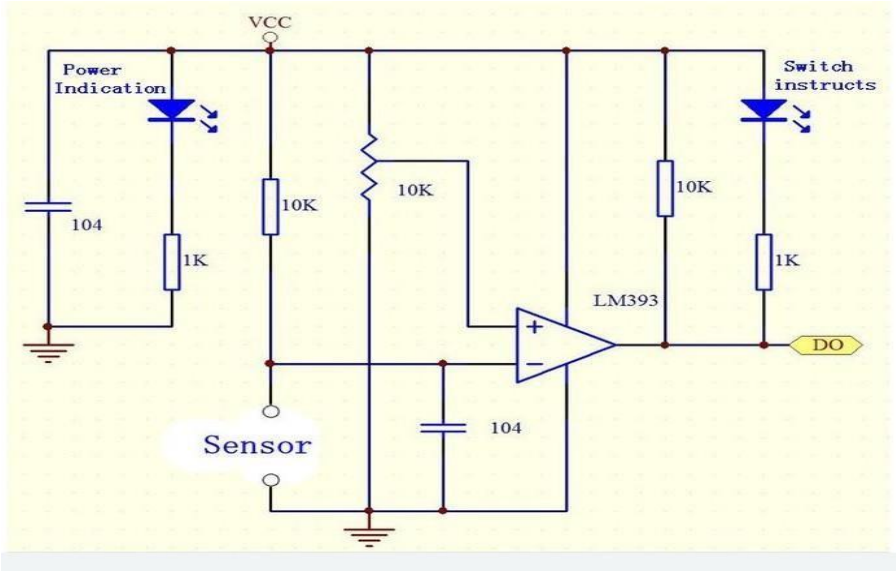
Specifications:

- On-board LM393 voltage comparator chip and infrared sensing probe.
- Support 5V/3.3V voltage input.
- On-board signal output indication, output effective signal is high level, and the same time the indicator light up, output signal can directly connect with microcontroller IO.
- Signal detection sensitivity can be adjusted.
- Reserved a line voltage compare circuit (P3 is leaded out).
- PCB size: 30(mm) x15(mm).



Pin Configuration :

- VCC
- Output
- Ground



How to test:

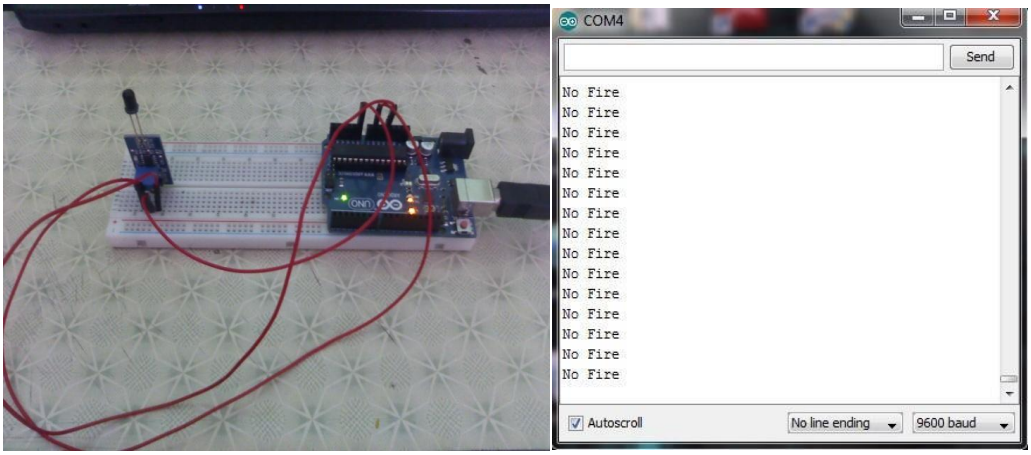
1. Connect your Arduino microcontroller to the computer.
2. Connect the VCC pin of your module to the to the 5V pin of your Arduino.
3. Connect the GND pin of your module to the GND pin of your Arduino.
4. Connect the Output pin of your module to the A0 pin of you Arduino.
5. Enter this program to your Arduino Integrated Development Environment (IDE):

```
void setup()
{
  Serial.begin(9600);
}
void loop()
{
  if (analogRead(A0) < 250) Serial.println("No Fire");
  else Serial.println("There's a Fire!");
   Delay(100);
    }
```

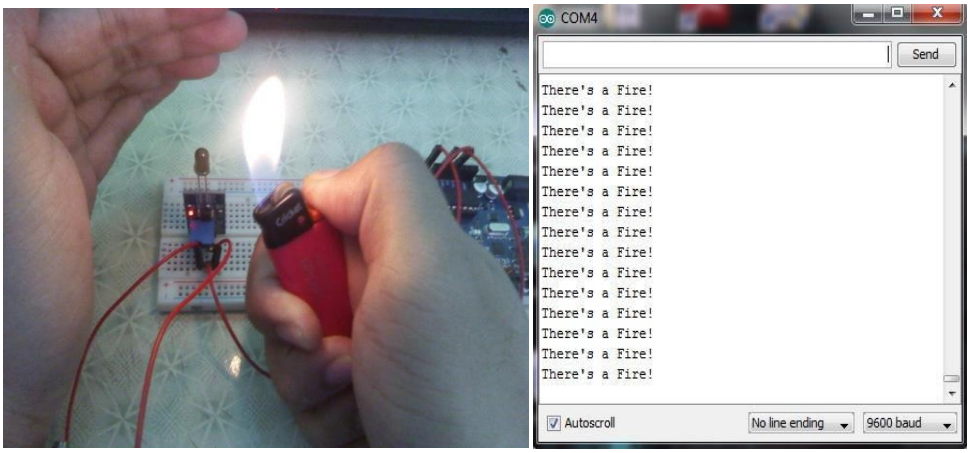
1. Click the Upload Button
2. Lastly, click the Serial Monitor button.

Testing Results:

When there is no fire:



When exposed to the fire of an ignited lighter:



8. FEATURES :

Real-time Tracking and Monitoring

1. Instantaneous attendance capture

2. Up-to-the-minute presence information
3. Real-time tracking of individuals within a facility
4. Automated alerts for unauthorized access or suspicious activity
5. Remote monitoring of attendance data

Automated Data Collection and Management

1. Automatic recording of attendance data
2. Elimination of manual entry and associated errors
3. Centralized database for easy data storage and retrieval
4. Time-stamped records for accurate tracking of arrival and departure times
5. Automated generation of attendance reports
6. Integration with existing HR or student management systems

Enhanced Security and Access Control

- Restricted access to specific areas based on RFID credentials
- Prevention of unauthorized entry and buddy punching
- Enhanced security measures for sensitive areas
- Integration with surveillance systems for comprehensive security monitoring
- Control over entry and exit points within a facility

Improved Efficiency and Productivity

- Streamlined attendance tracking process
- Reduced administrative burden
- Increased accuracy and reliability of attendance data
- Improved time management and resource allocation
- Enhanced operational efficiency and productivity

Data Analysis and Reporting

- Detailed reports on attendance patterns and trends
- Identification of absenteeism and tardiness issues
- Analysis of employee or student performance based on attendance data
- Customized reports to meet specific needs
- Data-driven decision making for improved resource management

Other Benefits

- Cost-effectiveness compared to traditional attendance methods
- Increased accountability and transparency
- Improved communication between administrators and employees/students
- Enhanced customer service through efficient tracking of visitor access
- Scalability to accommodate growing organizations or institutions
- Integration with other technologies such as GPS and facial recognition

- Improved safety and security for students and employees
- Reduced administrative overhead for attendance management
- Enhanced data security through encryption and access controls
- Remote management and configuration of the system
- Real-time notifications and alerts for critical events
- Integration with mobile devices for convenient access
- Improved compliance with industry regulations and standards
- Enhanced brand image through the use of cutting-edge technology
- Improved customer satisfaction through efficient service delivery
- Increased employee engagement and motivation
- Reduced environmental impact through paperless operations
- Improved resource utilization through optimized scheduling
- Enhanced disaster preparedness through rapid evacuation procedures
- Improved workplace safety through real-time monitoring of employee locations
- Reduced risk of legal issues related to attendance disputes
- Enhanced data analytics capabilities for informed decision making
- Improved overall operational efficiency and effectiveness
- Increased return on investment through reduced costs and improved productivity

9. ADVANTAGES :

- **Increased Efficiency:**
  1. Automated Time Tracking: Eliminates manual time-keeping methods like punch cards, reducing administrative overhead.
  2. Reduced Payroll Errors: Accurate time and attendance data minimizes errors in payroll calculations, saving time and money.
  3. Improved Productivity: Real-time tracking can identify patterns of employee attendance and potential productivity issues.
- **Enhanced Security:**
  1. Access Control: Restricts access to specific areas within the workplace based on employee roles and permissions.
  2. Time and Attendance Verification: Prevents buddy punching and other forms of time theft.
- **Improved Data Management:**
  1. Centralized Data Storage: Stores all employee attendance data in a centralized database for easy access and analysis.
  2. Real-time Reporting: Generates real-time reports on employee attendance, tardiness, and absenteeism.
  3. Data-Driven Decisions: Provides valuable insights for workforce planning, scheduling, and performance management.

- **Cost Savings:**
  1. Reduced Labor Costs: Automates time-consuming manual tasks, freeing up human resources for more productive work.
  2. Improved Accuracy: Minimizes errors in payroll processing, reducing costs associated with corrections.
  3. Enhanced Security: Reduces the risk of theft and fraud, protecting company assets.
- **Employee Convenience:**
  1. Contactless and Quick: Employees simply tap their RFID cards at designated readers, making the process fast and convenient.
  2. Improved Accuracy: Eliminates the need for manual entry, reducing the risk of human error.
  3. Reduced Stress: Automates a routine task, reducing stress and improving employee morale.

For Students:

- **Improved Attendance Tracking:**
  1. Accurate and Reliable: Provides real-time and accurate attendance records, eliminating manual errors and discrepancies.
  2. Automated Tracking: Reduces the time and effort required for teachers to take attendance.
  3. Improved Data Analysis: Allows for easy analysis of attendance patterns, identifying trends and potential issues.
- **Enhanced Student Safety:**
  1. Real-time Tracking: Enables quick identification of students who are absent or late.
  2. Emergency Response: Can be integrated with emergency alert systems for faster response times in case of incidents.
- **Improved Classroom Management:**
  1. Streamlined Check-in/Check-out: Facilitates smooth entry and exit from classrooms.
  2. Improved Student Engagement: Encourages punctual and responsible attendance habits.
- **Administrative Efficiency:**
  1. Automated Reporting: Generates automated reports for administrators, teachers, and parents.
  2. Reduced Paperwork: Eliminates the need for manual attendance sheets and records.
  3. Improved Data Analysis: Provides valuable insights for school administrators to make informed decisions regarding student support and resource allocation.

General Advantages:

- Increased Accuracy: Eliminates human error associated with manual data entry.
- Improved Efficiency: Automates time-consuming tasks, freeing up time for other important activities.
- Enhanced Security: Reduces the risk of fraud and unauthorized access.
- Improved Data Management: Provides access to real-time data for better decision-making.

- Cost-Effectiveness: Reduces administrative costs and improves overall operational efficiency.
- Increased Transparency: Provides a clear and auditable record of attendance.
- Improved Accountability: Encourages timely and responsible attendance.

By leveraging the advantages of RFID technology, organizations can streamline their attendance management processes, improve efficiency, enhance security, and gain valuable insights into workforce or student behavior.

9. DISADVANTAGES:

- Privacy Concerns:
    1. Constant Monitoring: Employees may feel constantly monitored, which can lead to stress and decreased job satisfaction.
    2. Data Privacy: Concerns about the security and privacy of personal data stored in the system.
- Technical Issues:
    1. System Malfunctions: Technical glitches, such as reader malfunctions or tag failures, can disrupt the attendance process and cause frustration.
    2. Interference: RFID signals can be susceptible to interference from other electronic devices, leading to inaccurate readings.
- Cost:
    1. Initial Investment: The initial cost of implementing an RFID system can be significant, including hardware (readers, tags), software, and installation.
    2. Maintenance Costs: Ongoing maintenance and system upgrades can also incur costs.
- Dependence on Technology:
    1. System Reliability: The system's effectiveness relies heavily on the reliability of the technology and the maintenance of the equipment.
    2. Power Outages: Power outages can disrupt the system, causing delays and potential data loss.

For Students:

- Privacy Concerns:
    1. Student Monitoring: Students may feel their privacy is being compromised by constant tracking.
    2. Data Security: Concerns about the security and privacy of student data.
- Technical Issues:
    1. Tag Loss or Damage: Lost or damaged RFID tags can cause attendance issues and inconvenience for students.

2. Reader Malfunctions: Technical problems with the readers can disrupt the attendance process and cause delays.

- Cost:

  1. Implementation Costs: The cost of implementing and maintaining the system can be a burden for schools with limited budgets.

- Equity and Accessibility:

  1. Cost of Tags: Students may need to bear the cost of RFID tags, which can create inequities.
  2. Accessibility for Students with Disabilities: Some students with disabilities may face challenges with using RFID technology.

General Disadvantages:

- Security Vulnerabilities:

  1. Data Breaches: The system is vulnerable to data breaches, which could expose sensitive employee or student information.
  2. Unauthorized Access: Potential for unauthorized access to the system and data.

- Lack of Flexibility:

  1. Limited Customization: May not be easily adaptable to specific organizational needs or workflows.
  2. Difficulty in Modifications: Making changes to the system can be complex and time-consuming.

- Environmental Impact:

  1. E-waste: Disposal of old RFID tags and readers can contribute to electronic waste.

It's important to carefully weigh the potential advantages and disadvantages before implementing an RFID-enabled attendance system. A thorough cost-benefit analysis and careful consideration of privacy and security concerns are crucial for successful implementation.

10. ARDUINO CODE:

```
int f1=A0;
int f2=6;
int f=9;
int led2=7;
String id;
int b=0;
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
// Set the LCD address to 0x27 for a 16 chars and 2 line display
LiquidCrystal_I2C lcd(0x27, 16, 2);
void setup() {
  // put your setup code here, to run once:
  Serial.begin(9600);
    pinMode(f,INPUT);
  pinMode(f1,INPUT);
  pinMode(f2,INPUT);
  pinMode(led2,OUTPUT);
    lcd.init();
  // Turn on the blacklight and print a message.
  lcd.backlight();
  lcd.print("Hello, world!");
}

void loop() {
  // put your main code here, to run repeatedly:
   if (Serial.available() > 0)
```

```
{
  id = Serial.readString();
  Serial.println(id);
  delay(1000);

if(id=="150027371712")
{
    Serial.println("id ...1");
    lcd.clear();
     lcd.setCursor(0,0);
     lcd.print("SRINADH");
      lcd.setCursor(0,1);
  lcd.print(" ROLL NUM :4B0");
}// waits 15ms for the servo to reach the position
  if(id=="150027371712")
{
    Serial.println("id ...1");
    lcd.clear();
     lcd.setCursor(0,0);
     lcd.print("SRINADH");
      lcd.setCursor(0,1);
  lcd.print(" ROLL NUM :4B0");
   delay(5000);
//}// waits 15ms for the servo to reach the position
}
  if(id=="55001486A562")
{
    Serial.println("id ...2");
        //delay(5000);
```

```
    lcd.clear();
     lcd.setCursor(0,0);
     lcd.print("GANESH");
      lcd.setCursor(0,1);
  lcd.print(" ROLL NUM :4C7");
  delay(5000);
//}// waits 15ms for the servo to reach the position
}
}
int x= analogRead(f1);
Serial.println("gas");
Serial.println(x);
int y= digitalRead(f);
Serial.println("fir");
Serial.println(y);
  lcd.clear();
 lcd.print("gas:");
 lcd.print(x);
 delay(1000);
 lcd.clear();
 lcd.print("fir:");
 lcd.print(y);
delay(3000);
int z= digitalRead(f2);
Serial.println("vibration");
Serial.println(z);
  lcd.clear();
 lcd.print("vibration:");
 lcd.print(z);
```

```
  delay(1000);
 if(x>=580)
 {
   digitalWrite(led2,HIGH);
 }
 else
 {
   digitalWrite(led2,LOW);
 }
if(y==0)
 {
    digitalWrite(led2,HIGH);
}
else
{
  digitalWrite(led2,LOW);
}
if(z==0)
 {
    digitalWrite(led2,HIGH);
}
else
{
  digitalWrite(led2,LOW);
}
}
```

11. CONCLUSION:

In conclusion, RFID-enabled attendance systems offer a compelling solution for automating employee and student tracking, revolutionizing traditional methods. By leveraging the power of radio-frequency identification technology, these systems deliver significant advancements in efficiency, accuracy, and security.

For employees, RFID systems streamline time tracking, minimize payroll errors, and enhance workplace security. Automated attendance eliminates the need for manual data entry, reducing administrative burden and freeing up valuable human resources. Real-time data analysis provides valuable insights into workforce trends, enabling informed decision-making regarding staffing, scheduling, and resource allocation.

In educational settings, RFID systems enhance student safety, improve classroom management, and streamline administrative processes. Automated attendance tracking reduces teacher workload, improves data accuracy, and facilitates timely identification of student absences. These systems also contribute to a more secure and efficient learning environment.

While challenges such as initial investment costs, potential privacy concerns, and technological limitations exist, the benefits of RFID-enabled attendance systems are undeniable. Continuous advancements in RFID technology, coupled with robust data security measures and user-friendly interfaces, are paving the way for a more efficient and streamlined future for attendance management across various sectors. As technology continues to evolve, we can expect to see further integration of RFID systems with other technologies, such as artificial intelligence and the Internet of Things, leading to even more innovative and impactful solutions for attendance tracking and beyond.

Disclaimer: This information is for general knowledge and informational purposes only and does not constitute professional advice.

Note: This conclusion provides a general overview. Specific applications and conclusions may vary depending on the specific context and implementation of the RFID system.

11. REFERENCES:

[1] N. Amiyana and M. Alias, “Attendance and Access Control System Using RFID System”, (2011) .

[2] C. E. Geoggrey, “Automatic Access Control System using Student’s Identification Card based on RFID Technology”, (2012).

[3] D. M. Dobkin and S. M. Weigand, “Environmental Effects on RFID Tag Antennas”, California, Bulis Press,2010.

[4] T. S. Lim, S.C. Sim and M. M Mansor, “ RFID Based Attendance System” 2009 IEEE Symposium on Industrial Electronics and Applications (ISIEA 2009), Kuala Lumpur, Malaysia, October 4-6, 2009.

[5] Radio-frequency identification. Wikipedia (online). Wikimedia Foundation, Inc., 2009.

[6] Automated Time and Attendance System. Retrieved from https://www.gatekeepersolutions.com..

[7] L. Sandip, “RFID Sourcebook”, IBM Press, USA, (2005)ISBN: 0-13 185137-3 .

[8] D. R. Hancox, “Using RFID Technology to Enhance Corporate Effectiveness” ,April 2016

[9] EPC Generation 1 Tag Data Standards Version 1.1 Rev. 1.27, EPC Global Inc. Standard Specification, May 2005.

[10] MIT Auto-ID Center. Draft Protocol Specification for a 900 MHz Class of Radio Frequency Identification Tag, February 2003.

[11] K. Finkenzeller. RFID Handbook, John Wiley & Sons, 1999.

[12] D.R. Hush. A Comparison of Arbitration Algorithms for RFID. Technical report, Micron Communications, Inc., Boise, ID, 1997.

[13] D.R. Hush, C. Wood, Analysis of Tree Algorithms for RFID Arbitration, In Proceedings of IEEE International Symposium on Information Theory pg. 107, 1998.

[14] C. Law, K. Lee, and K. Y. Siu. Efficient memoryless protocol for tag identification, In Proceedings of DIALM, Boston, MA, Aug. 2000, pp. 75-84.

[15] J. Myung and W. Lee. Adaptive Binary Splitting: A RFID Tag Collision Arbitration Protocol for Tag Identification, accepted for publication in ACM MONET, 2006.

[16] V. Namboodiri and L. Gao. Energy-Aware Tag Anti-Collision Protocols for RFID Systems, ECE Tech. Report, 2006. Available at www.people.umass.edu/ vnambood/publications.html.

[17] F. Zhou, C. Chen, D. Jing, C. Huang and H. Min. Evaluating and optimizing power-consumption of anti-collision protocols for applications in RFID systems, Auto-ID Labs Tech. Report AUTOIDLABS-WPSWNET-014.

[18] Radio frequency identification - a basic primer. White Paper, AIM Ine WP-98/002R2, August 2001.

[19] D. Engels and S. Sarma. The Reader Collision Problem. IEEE International Conference on Systems, Man and Cybernetics, 3:6, 2002.

[20] A. Sahoo, S. Iyer, and N. Bhandari. Improving rfid system to read tags efficiently, http://www.it.iitb.ac.in/research/techreport/reports/18.pdf, 2006.

[21] H. Vogt. Multiple object identification with passive RFID tags. In IEEE International Conference on Systems, Man and Cybernetics (SMC '02), Oct. 2002.

[22] F. Zhou, C. Chen, D. Jin, C. Huang, and H. Min. Evaluating and optimizing power consumption of anti-collision protocols for applications in RFID systems. In ISLPED '04: Proceedings of the 2004 international symposium on Low power electronics and design, pages 357-362, New York, NY, USA, 2004. ACM Press.

[23] J. Waldrop, D. Engels, and S. Sarma. Colorwave : An Anticollision Algorithm for the Reader Collision Problem. IEEE International Conference on Communications, 2:1206 - 1210, 2003.